Sinais e Sistemas para Engenharia

MÓDULO 3 - INTRODUÇÃO AOS SISTEMAS

VERSÃO OCTAVE

PROFESSOR LUCIANO NEVES DA FONSECA

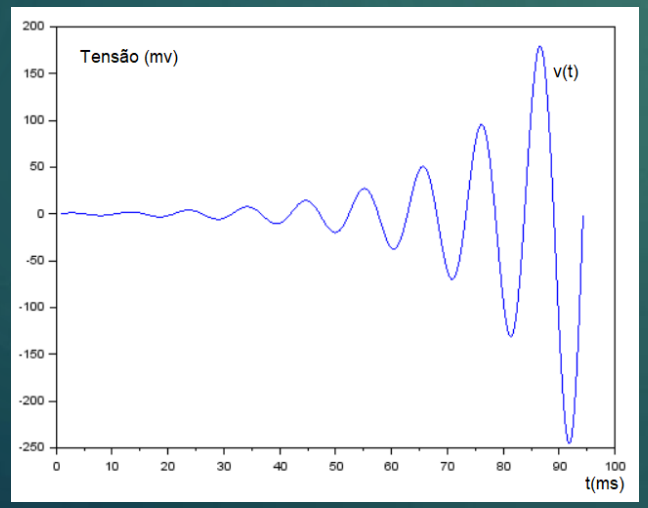
Entidade que processa um ou mais sinais de entrada, produzindo um sinal de saída.

Exemplo 1: Sistema eletrônico que compreende um amplificador/equalizador de som com caixa acústica, composto por um circuito eletrônico com resistores, capacitores, amplificadores operacionais etc.

O sinal de entrada pode ser uma música que seu telefone celular manda para o amplificador. O sistema é o amplificador/equalizador e a caixa de som. O sinal de saída do sistema seria a pressão acústica em mPa na caixa de som, do sinal de áudio amplificado e equalizado.

Sinal de Entrada

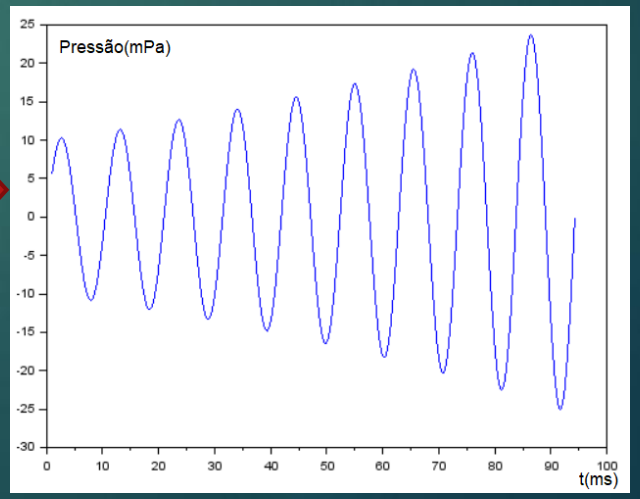
$y_{in} = f_i(t)$



Sistema
(componentes eletrônicos)

Amplificador com caixa de som

Sinal de saída



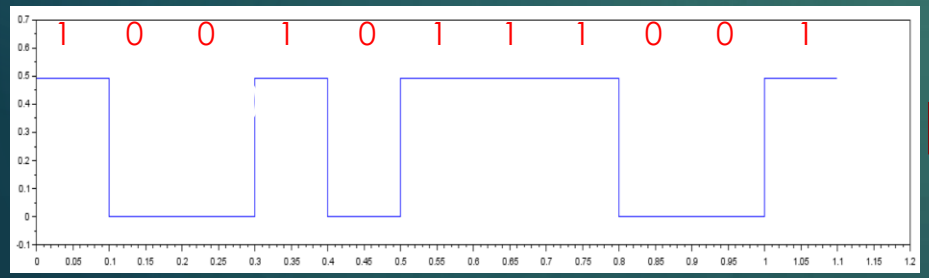
$y_{out} = f_o(t)$

Entidade que processa um ou mais sinais de entrada, produzindo um sinal de saída.

Exemplo 2: Sistema analógico que modela o meio de transmissão de um sinal

O sinal de entrada é gerado, por exemplo, pelo seu modem. O sistema é o meio de transmissão (no caso o ar), que irá alterar (distorcer!) o sinal de entrada. O sinal de saída é o que chega no seu laptop

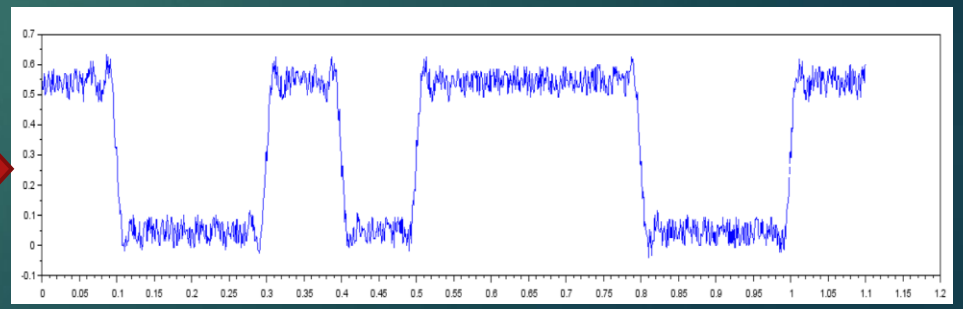
Sinal de Entrada



Sistema
(meio de transmissão)

Meio de Transmissão

Sinal de saída

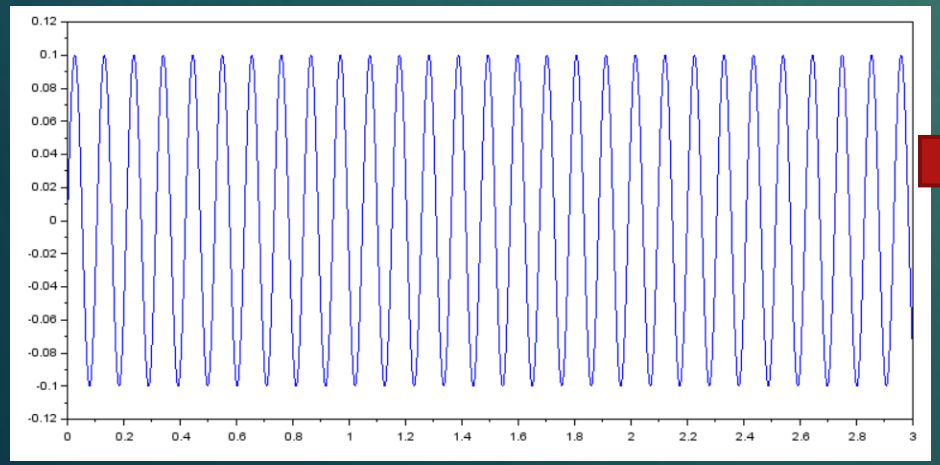


Entidade que processa um ou mais sinais de entrada, produzindo um sinal de saída.

Exemplo 3 Sistema analógico que modela a suspensão de um automóvel, com amortecedores, molas e massas

O sinal de entrada é por exemplo o perfil de uma estrada de terra, onde há um oscilação de 10cm de amplitude. O sistema é a suspensão do automóvel que absorve este movimento. O sinal de saída é o movimento efetivo da cabine do automóvel, que reflete a oscilação da estrada, mas bem amortecido.

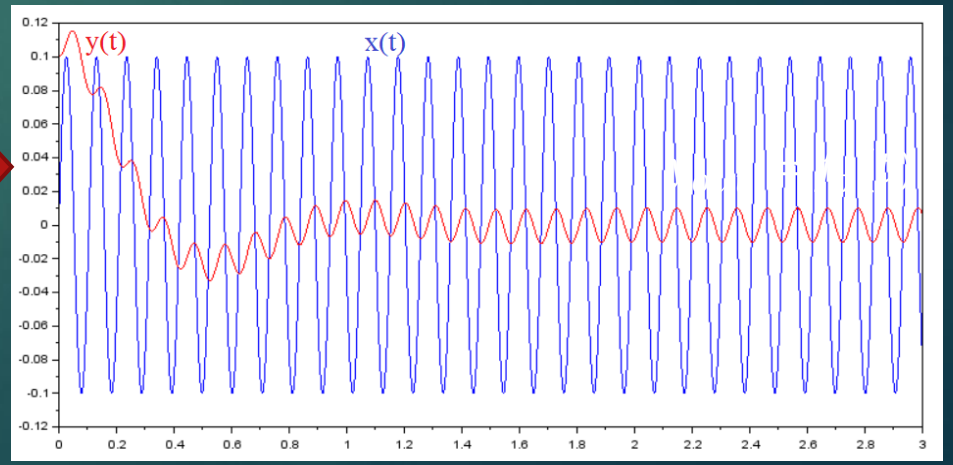
Sinal de Entrada



Sistema
(massa, mola amortecedor)

Suspensão de
um automóvel

Sinal de saída



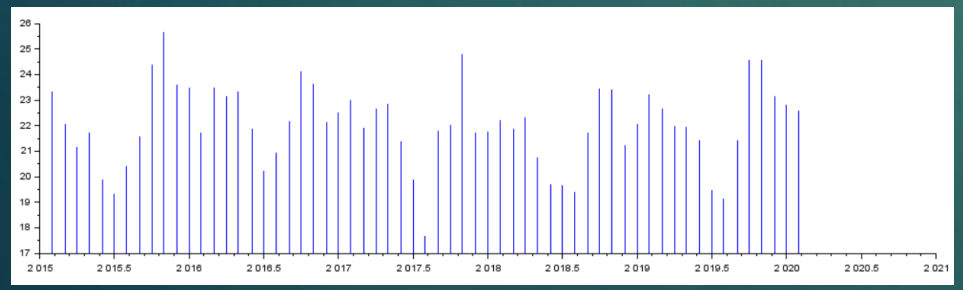
Entidade que processa um ou mais sinais de entrada, produzindo um sinal de saída.

Exemplo 4: Sistema Digital que tira a média das 6 últimas amostras

O sinal de entrada é o sinal discreto, 1 amostra por mês, com a temperatura do DF desde 2015. O sistema é um algoritmo que calcula a média das 6 últimas amostras. A saída é o sinal discreto que contém a média dos últimos 6 meses.

$y_{in} = f_i[n]$

Sinal de Entrada

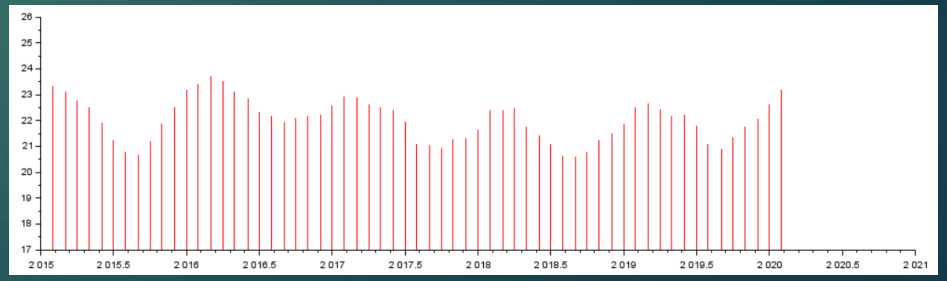


Sistema
(média 6 amostras)

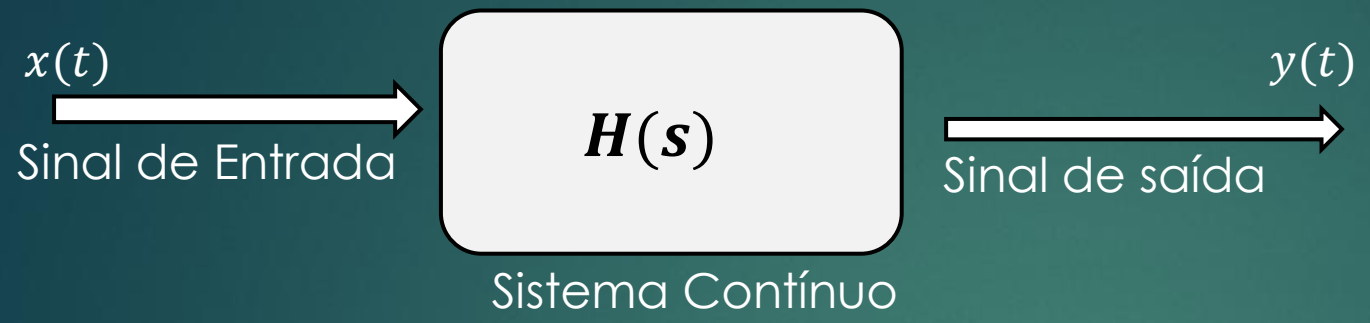
Algoritmo

Sinal de saída

$y_{out} = f_o[n]$

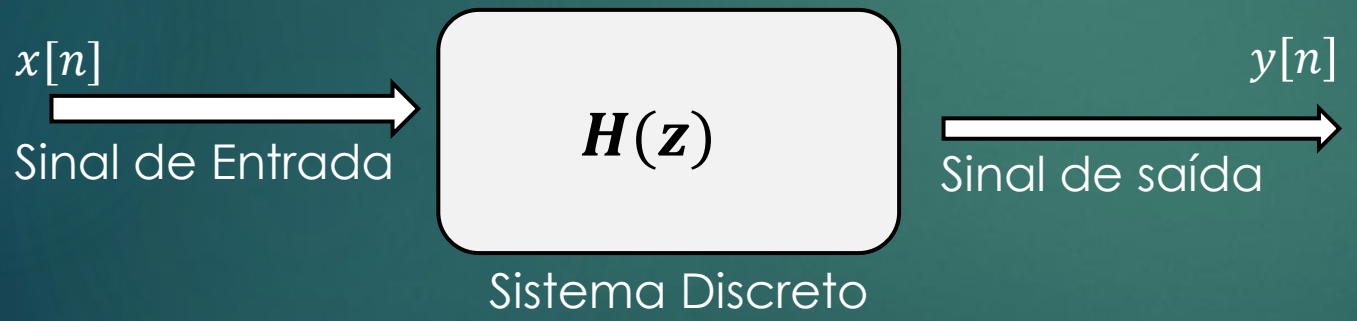


Sistemas de tempo contínuo e de tempo discreto



$$x(t) \xRightarrow{H(s)} y(t)$$

Um sistema de tempo contínuo H(s), implementado por exemplo por um hardware de circuito eletrônico com resistores, capacitores e circuitos integrados, transforma o sinal contínuo de entrada x(t) em um sinal de saída y(t)



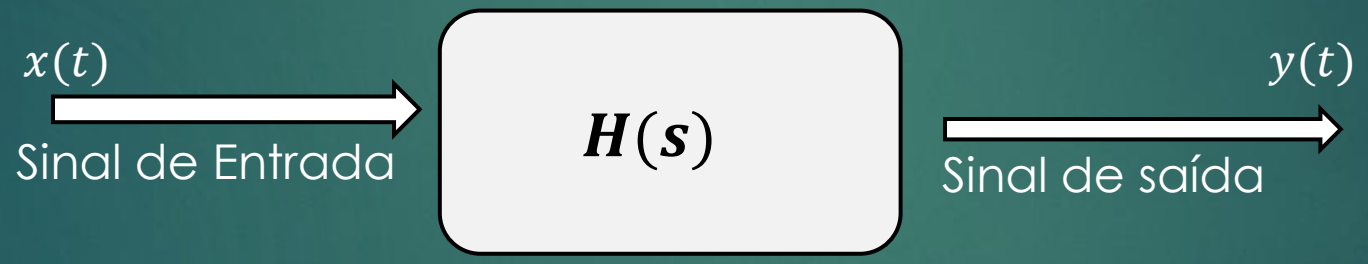
$$x[n] \xRightarrow{H(z)} y[n]$$

Um sistema de tempo discreto H(z), implementado por exemplo por um algoritmo em um computador, transforma (mapeia) a sequencia de números discretos de entrada x[n] em uma outra sequência de número discretos y[n] de saída.

Sistemas SISO Single-input single-output

Entrada única- Saída única

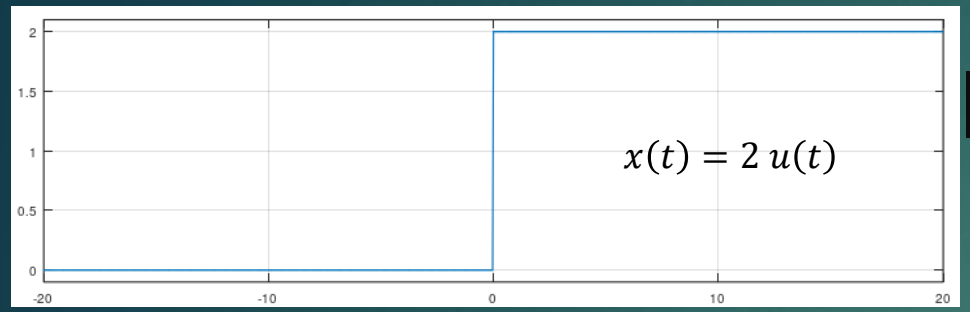
- Um sistema SISO de tempo contínuo/discreto aceita um único sinal de entrada e retorna um único sinal de saída.



Sistemas SISO Entrada única- Saída única

Exemplo: $x(t) \xRightarrow{5\,t\,x(t-3)\,-\,15} y(t)$

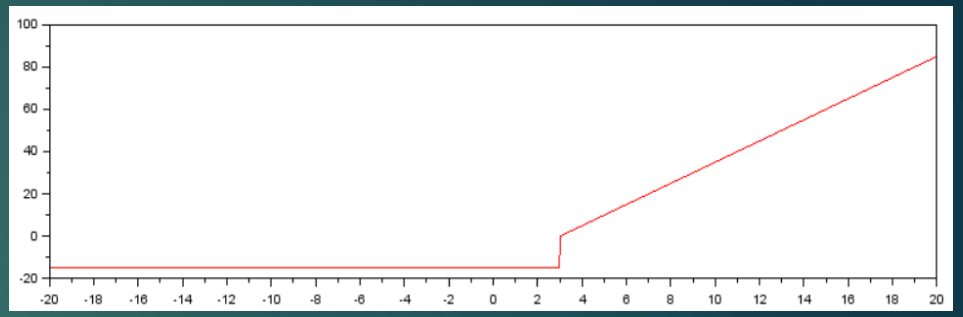
```
>> t=linspace(-20,20,1001);
>> x = @(t) 2*f_u(t);
>> plot(t,x(t))
```



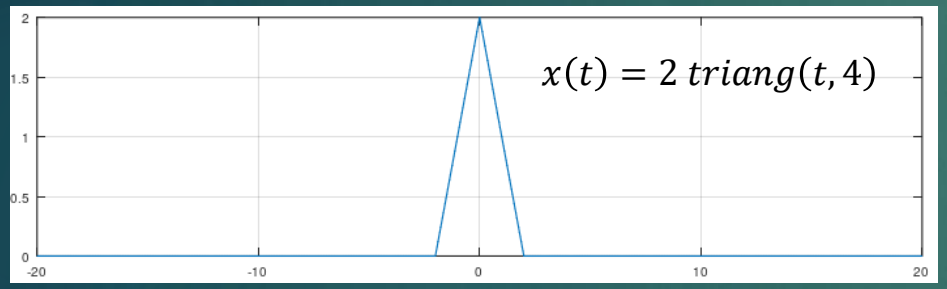

```
>> y = @(t,x) 5*t.*x(t-3)-15;
```

$x(t) \xRightarrow{5\,t\,x(t-3)\,-\,15} y(t)$

```
>> plot(t,y(t,x),'r');
```



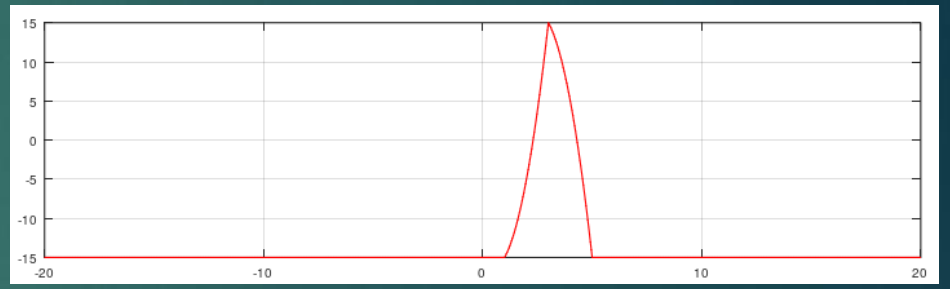
```
>> x = @(t) 2*f_pt(t,4);
>> plot(t,x(t))
```



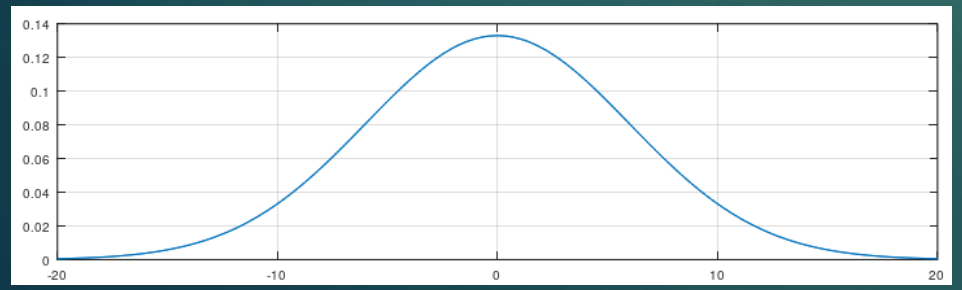

```
>> y = @(t,x) 5*t.*x(t-3)-15;
```

$x(t) \xRightarrow{5\,t\,x(t-3)\,-\,15} y(t)$

```
>> plot(t,y(t,x),'r');
```



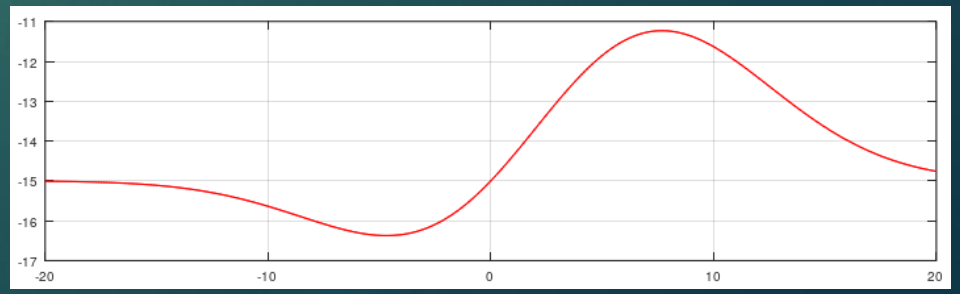
```
>> x = @(t) 2*f_pg(t,0,6);
>> plot(t,x(t))
```



```
>> y = @(t,x) 5*t.*x(t-3)-15;
```

$x(t) \xRightarrow{5\,t\,x(t-3)\,-\,15} y(t)$

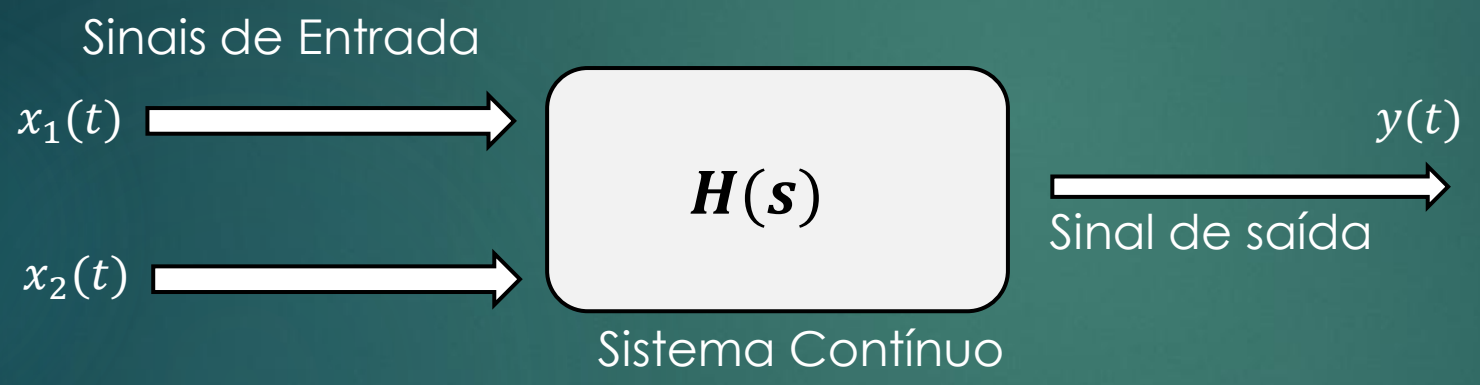
```
>> plot(t,y(t,x),'r');
```



Sistemas MISO Multiple-input single-output

Múltiplas Entradas - Saída única

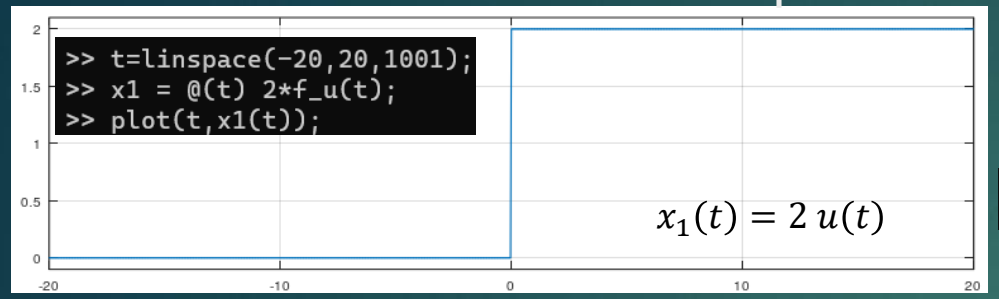
- Sistemas MISO (Multiple-input single-output) aceitam múltiplas entradas, mas retornam umsaída única.



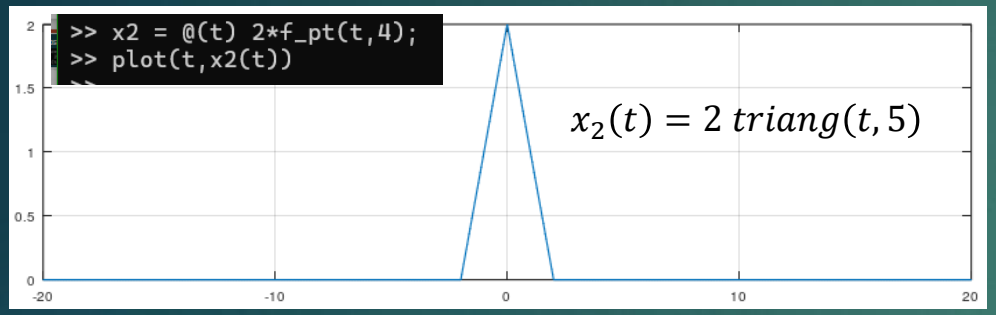
Sistemas MISO Múltiplas Entradas - Saída única

Exemplo:

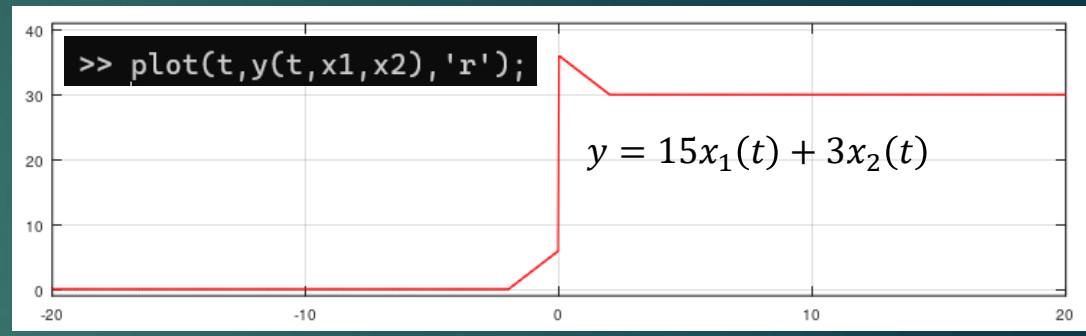
$$\begin{matrix} x_1(t) \\ x_2(t) \end{matrix} \xrightarrow{15x_1(t)+3x_2(t)} y(t)$$

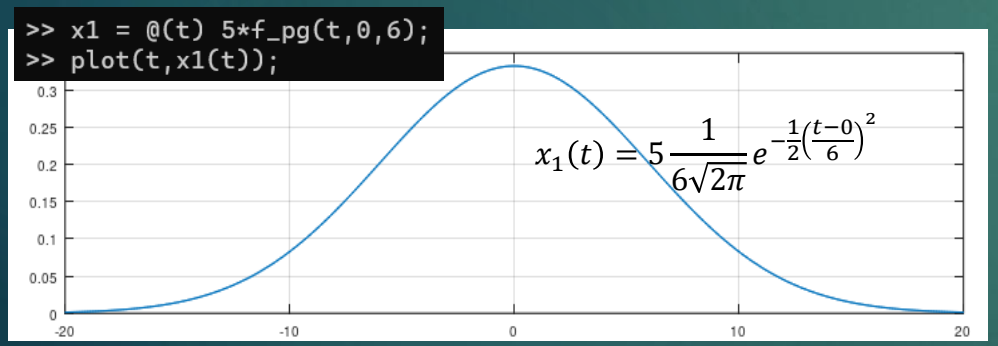



```
>> y = @(t,x1,x2) 15*x1(t)+2*x2(t);
```

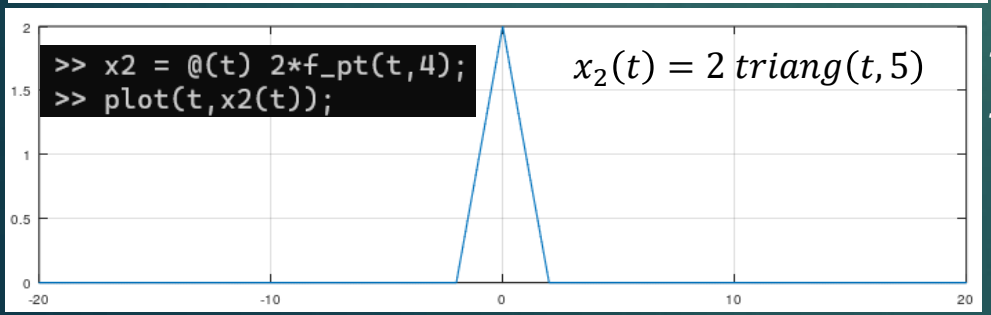


$$\begin{matrix} x_1(t) \\ x_2(t) \end{matrix} \xrightarrow{15x_1(t)+3x_2(t)} y(t)$$

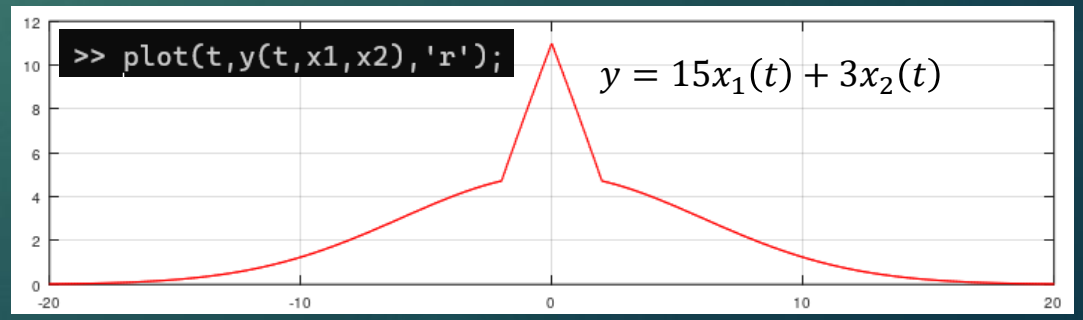





```
>> y = @(t,x1,x2) 15*x1(t)+2*x2(t);
```



$$\begin{matrix} x_1(t) \\ x_2(t) \end{matrix} \xrightarrow{15x_1(t)+3x_2(t)} y(t)$$



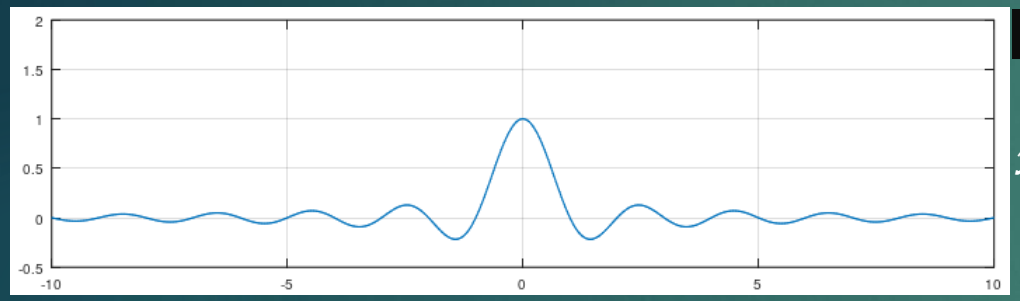
Sistemas Elementares

- 1) Escala de Magnitude

$$y(t) = k\,x(t)$$

$x(t)$ → Sinal de Entrada → $\boldsymbol{H(s)}$ → $y(t)$ → Sinal de saída

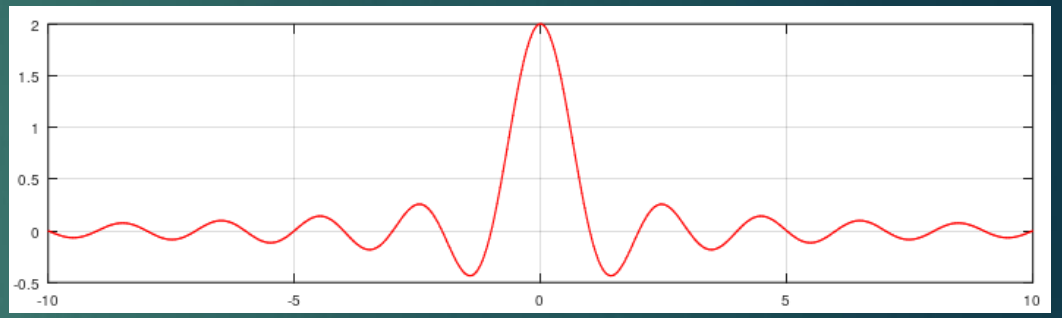
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_sc(t);
>> plot(t,x(t));
```



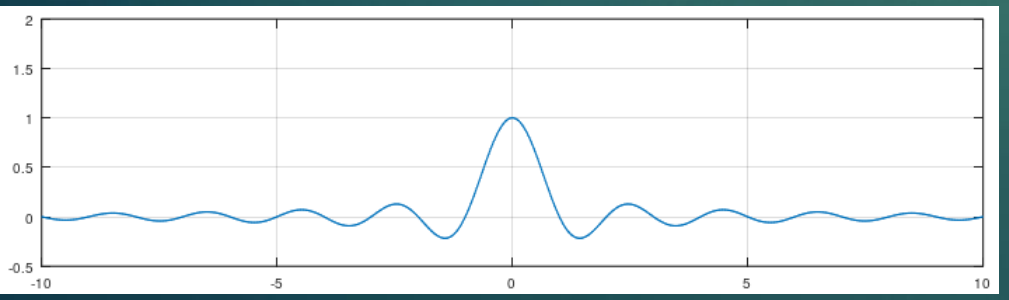

```
>> y1 = @(t,x) 2*x(t);
```

$$x(t) \xRightarrow{y(t)=2x(t)} y(t)$$

```
>> plot(t,y1(t,x),'r');
```



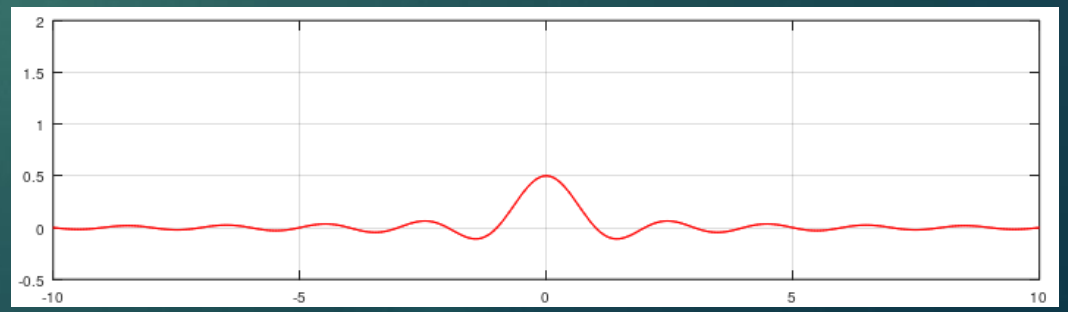

```
>> x = @(t) f_sc(t);
>> plot(t,x(t));
```



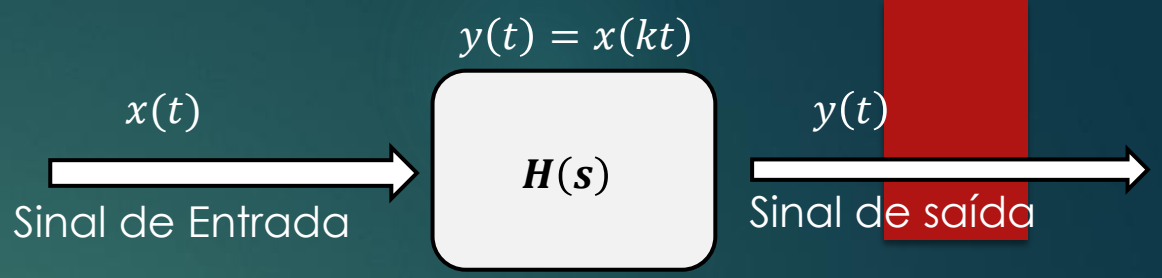

```
>> y2 = @(t,x) 0.5*x(t);
```

$$x(t) \xRightarrow{y(t)=\left(\frac{1}{2}\right)x(t)} y(t)$$

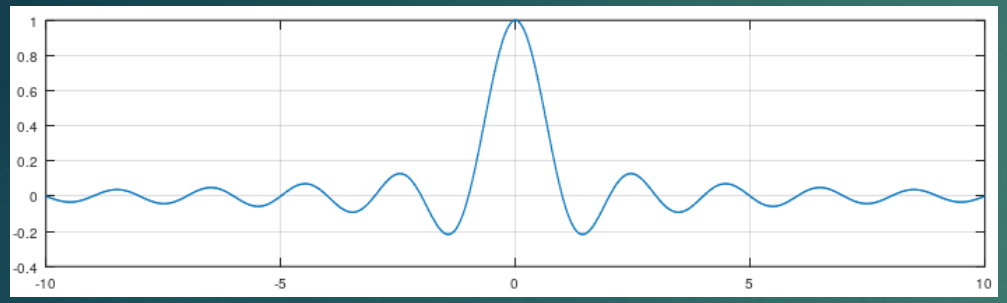
```
>> plot(t,y2(t,x),'r');
```



2) Escala no tempo



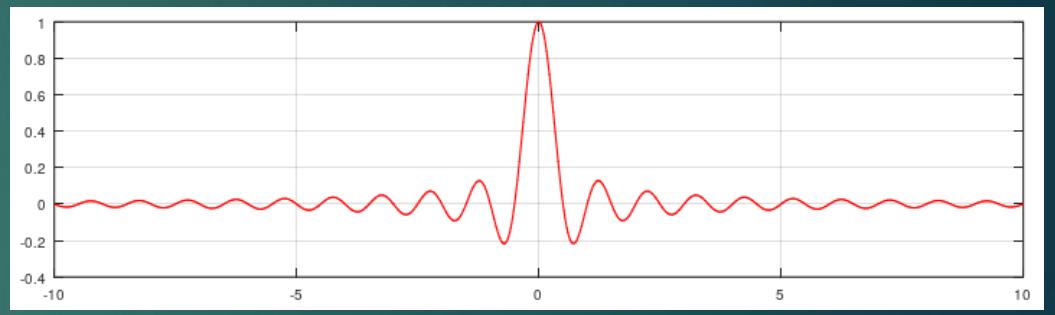

```
>> t=linspace(-10,10,1001);
>> x = @(t) f_sc(t);
>> plot(t,x(t));
```



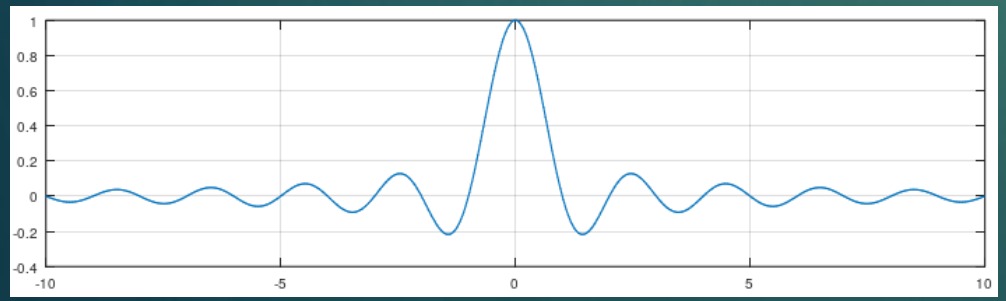
```
>> y1 = @(t,x) x(2*t);
```

$$x(t) \xRightarrow{y(t)=x(2\,t)} y(t)$$

```
>> plot(t,y1(t,x),'r');
```



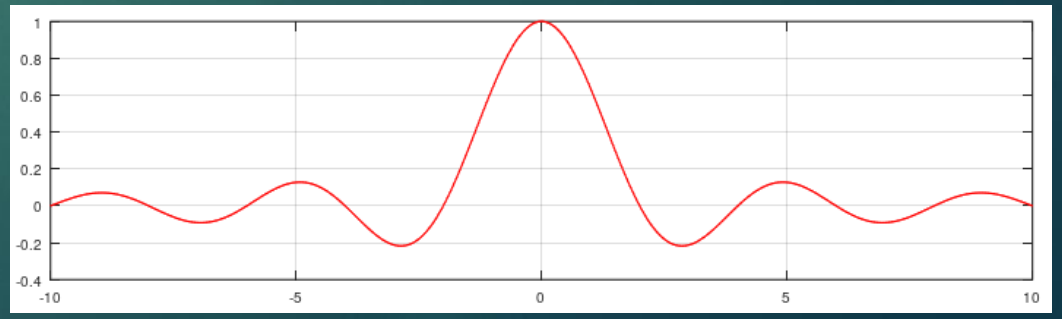
```
>> x = @(t) f_sc(t);
>> plot(t,x(t));
```



```
>> y2 = @(t,x) x(0.5*t);
```

$$x(t) \xRightarrow{y(t)=x(0.5\,t)} y(t)$$

```
>> plot(t,y2(t,x),'r');
```



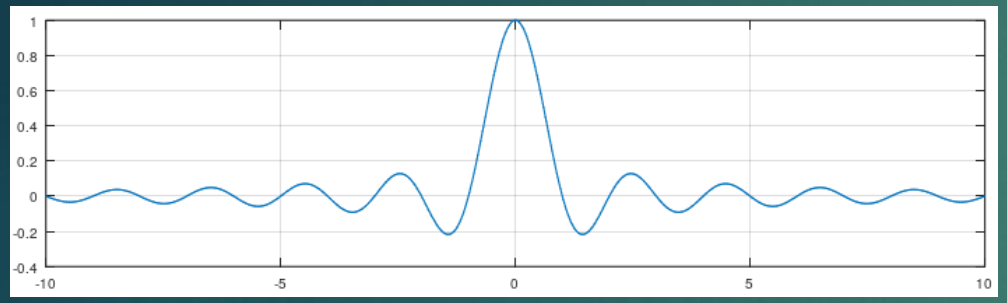
3) Deslocamento no tempo

$$y(t) = x(t + t_o)$$

$x(t)$ → Sinal de Entrada → $\boldsymbol{H(s)}$ → $y(t)$ → Sinal de saída

Sistema Contínuo

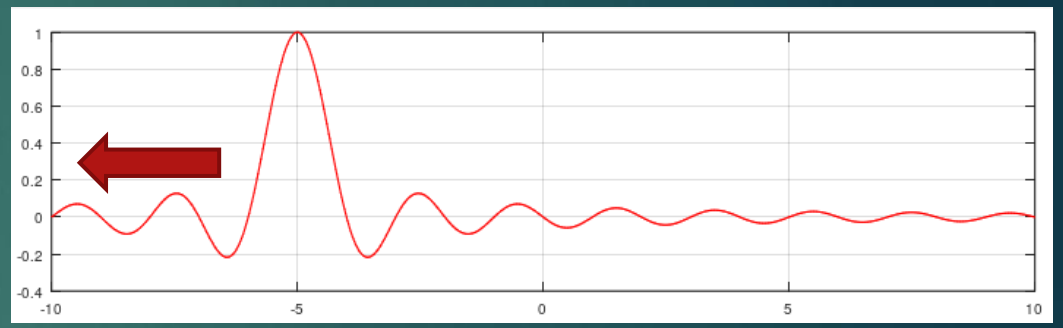
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_sc(t);
>> plot(t,x(t));
```



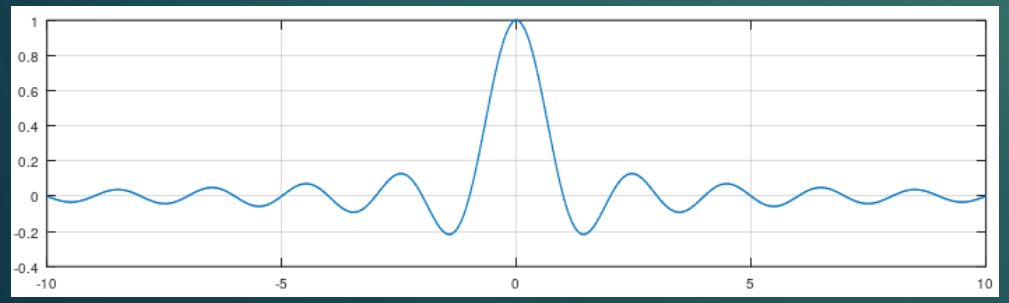
```
>> y1 = @(t,x) x(t+5);
```

$$x(t) \xRightarrow{y(t)=x(t+5)} y(t)$$

```
>> plot(t,y1(t,x),'r');
```



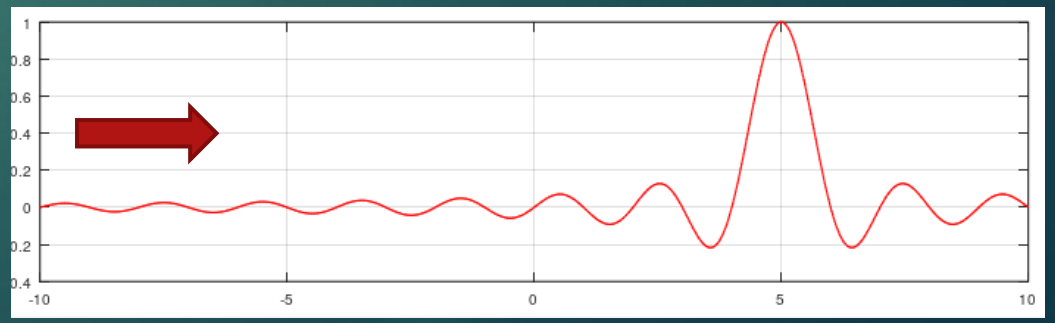
```
>> x = @(t) f_sc(t);
>> plot(t,x(t));
```



```
>> y2 = @(t,x) x(t-5);
```

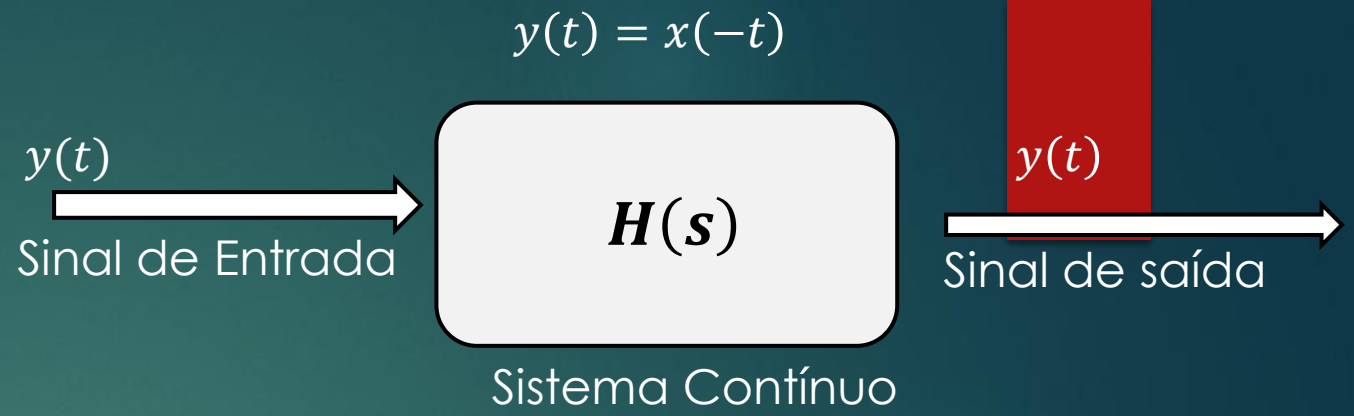
$$x(t) \xRightarrow{y(t)=x(t-5)} y(t)$$

```
>> plot(t,y2(t,x),'r');
```

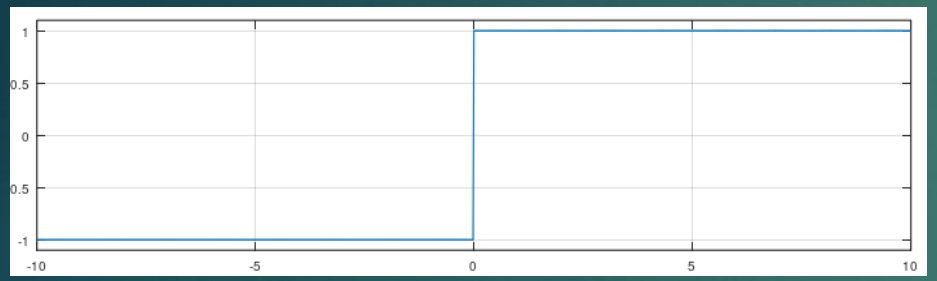


4) Inversão no tempo

$$y(t) = x(-t)$$



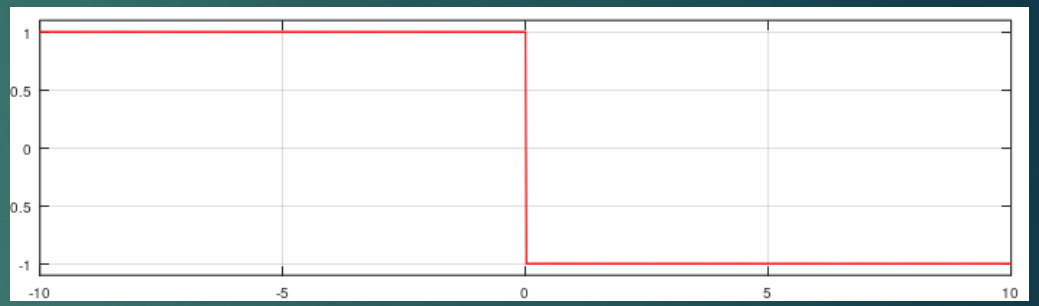

```
>> t=linspace(-10,10,1001);
>> x = @(t) f_s(t);
>> plot(t,x(t));
```



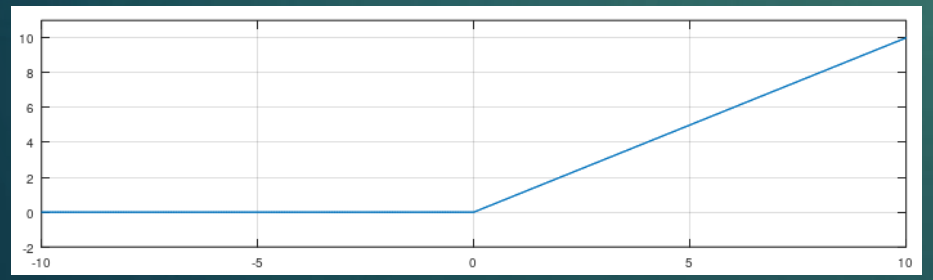
```
>> y = @(t,x) x(-t);
```

$$x(t) \xRightarrow{y(t)=x(-t)} y(t)$$

```
>> plot(t,y(t,x),'r');
```



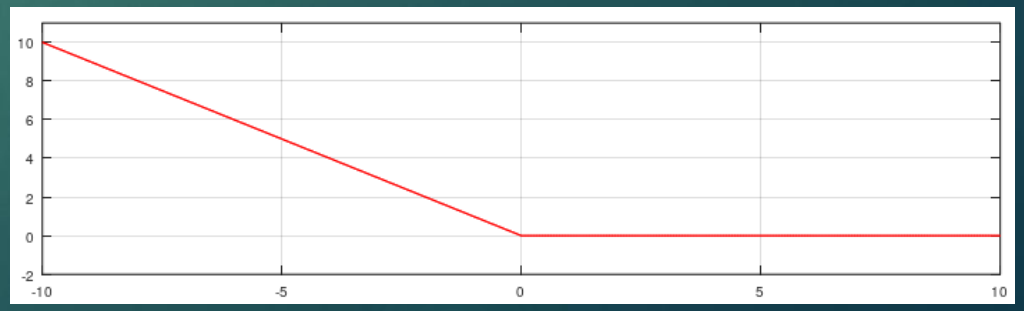
```
>> x = @(t) f_r(t);
>> plot(t,x(t));
```



```
>> y = @(t,x) x(-t);
```

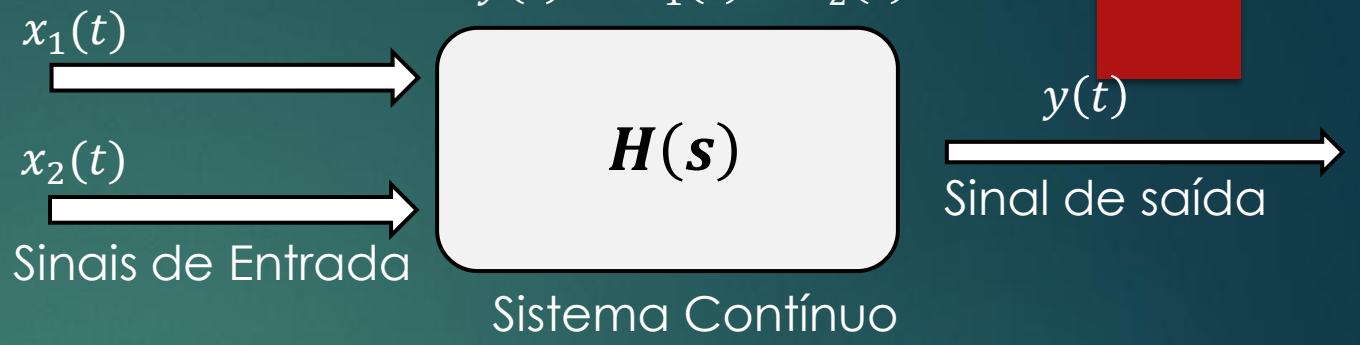
$$x(t) \xRightarrow{y(t)=x(-t)} y(t)$$

```
>> plot(t,y(t,x),'r');
```

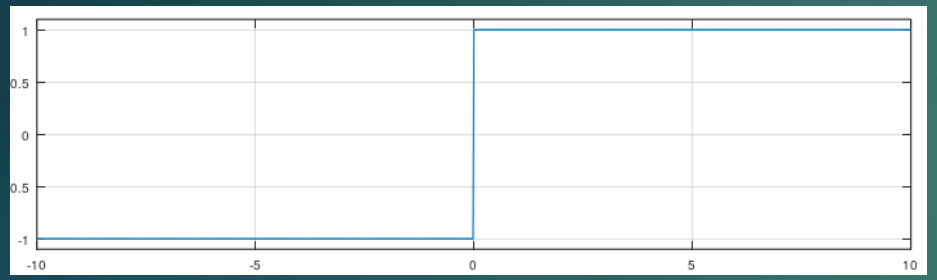


5) Adição de entradas

$$y(t) = x_1(t) + x_2(t)$$



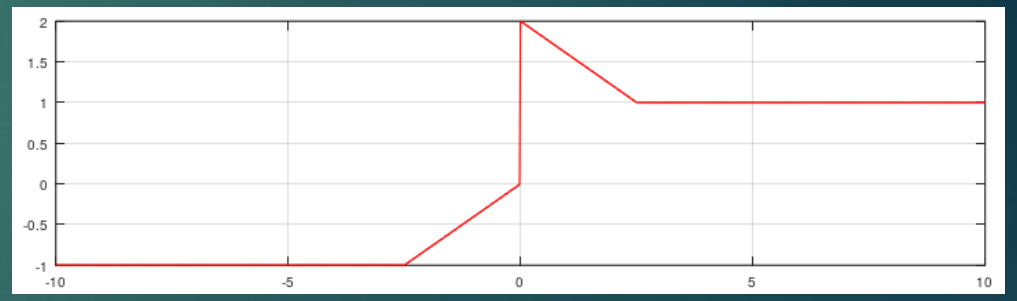

```
>> t=linspace(-10,10,1001);
>> x1 = @(t) f_s(t);
>> plot(t,x1(t));
```



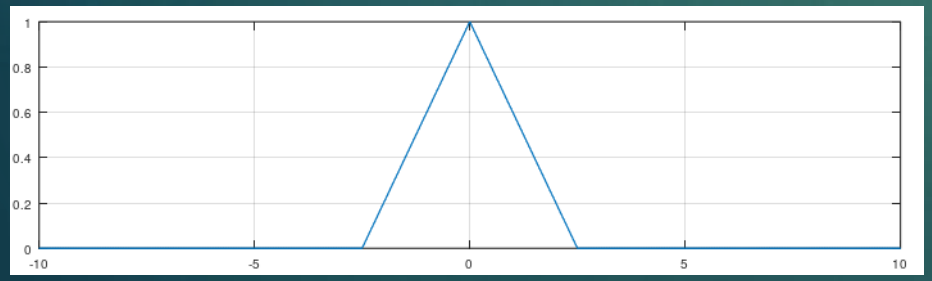
```
>> y = @(t,x1,x2) x1(t)+x2(t);
```

$$x_1(t),\ x_2(t) \xrightarrow{y(t)=x_1(t)+x_2(t)} y(t)$$

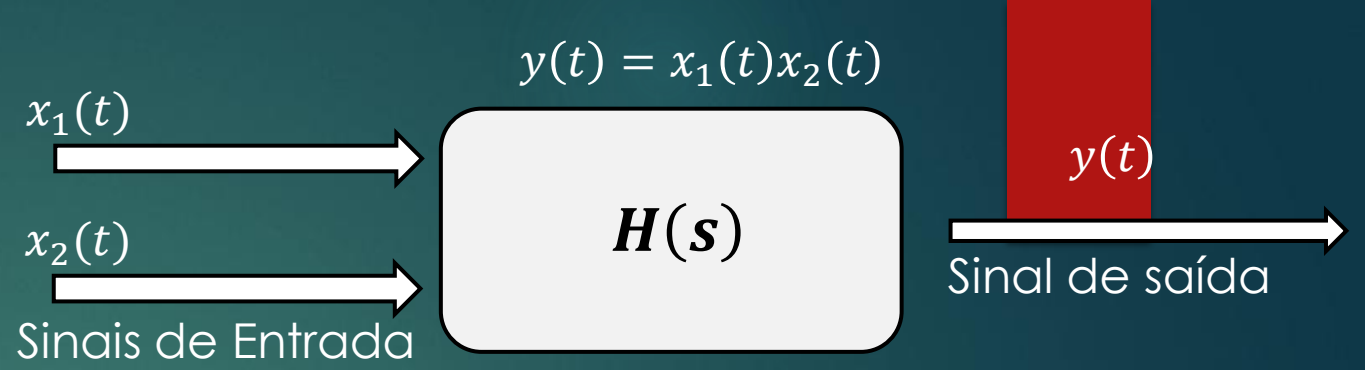
```
>> plot(t,y(t,x1,x2),'r');
```



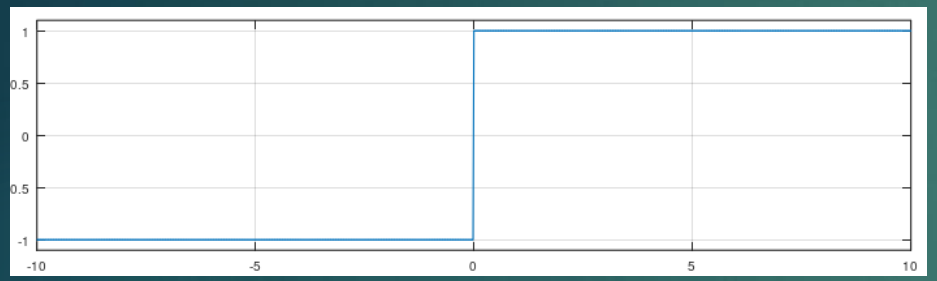
```
>> x2 = @(t) f_pt(t,5);
>> plot(t,x2(t));
```



6) Multiplicação de entradas



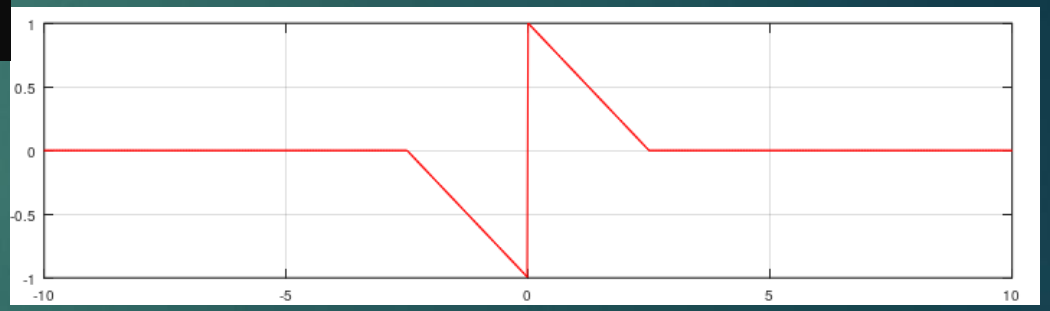

```
>> t=linspace(-10,10,1001);
>> x1 = @(t) f_s(t);
>> plot(t,x1(t));
```



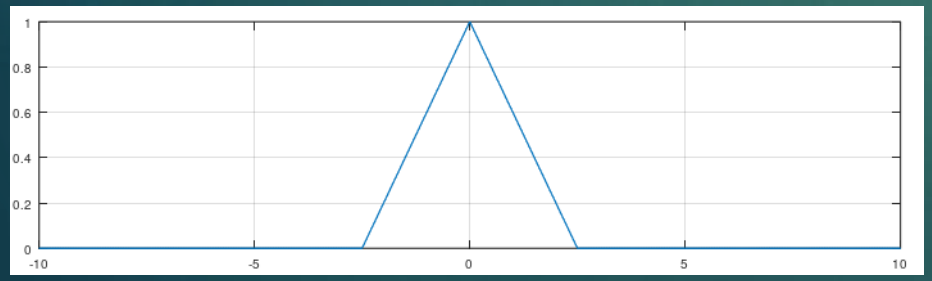
```
>> y = @(t,x1,x2) x1(t).*x2(t);
```

$$x_1(t),\ x_2(t) \xRightarrow{y(t)=x_1(t)x_2(t)} y(t)$$

```
>> plot(t,y(t,x1,x2),'r');
```



```
>> x2 = @(t) f_pt(t,5);
>> plot(t,x2(t));
```



7) Sistema diferenciador

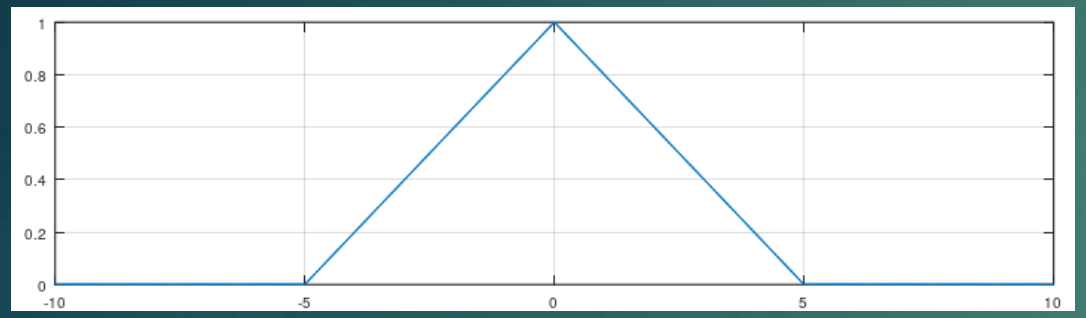
$$y(t) = \frac{dx(t)}{dt}$$

$x(t)$ → Sinal de Entrada

$$\boldsymbol{H(s) = s}$$

$y(t)$ → Sinal de saída

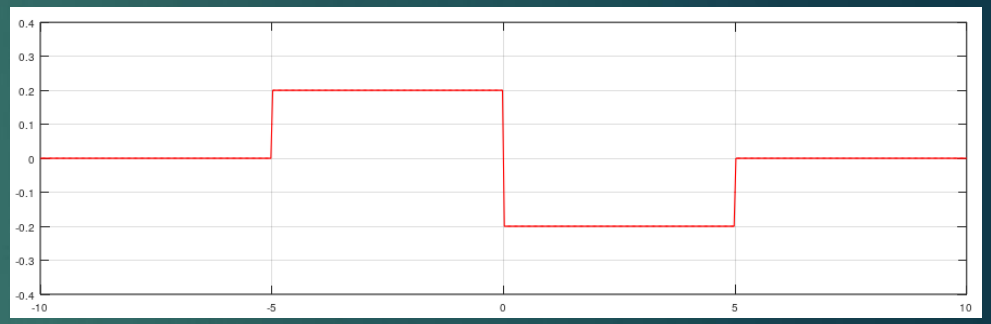
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_pt(t,10);
>> plot(t,x(t));
```




```
function y_out=dx(t,x)
```

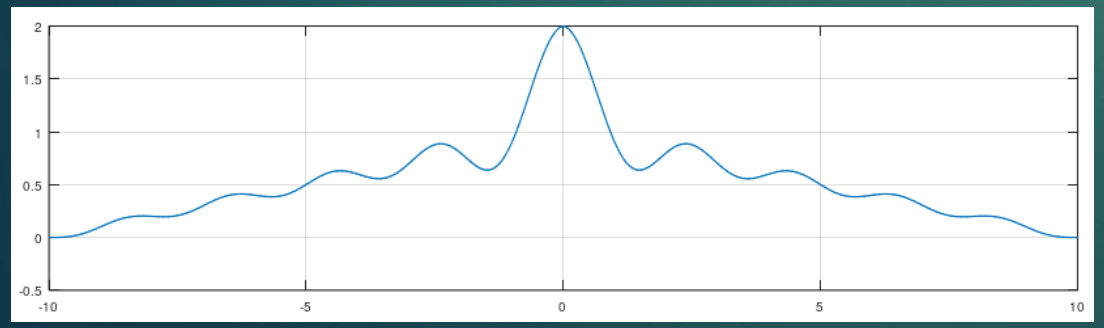
$$x(t) \xrightarrow{y(t)=\frac{dx(t)}{dt}} y(t)$$

```
>> plot(t,dx(t,x),'r')
```




```
function y_out=dx(t,x)
    N=length(t);
    y_out(1)=(x(t(2))-x(t(1)))/(t(2)-t(1));
    for(n=2:N-1)
        y_out(n)=(x(t(n+1))-x(t(n-1)))/(t(n+1)-t(n-1));
    end
    y_out(N)=(x(t(N))-x(t(N-1)))/(t(N)-t(N-1));
endfunction
```

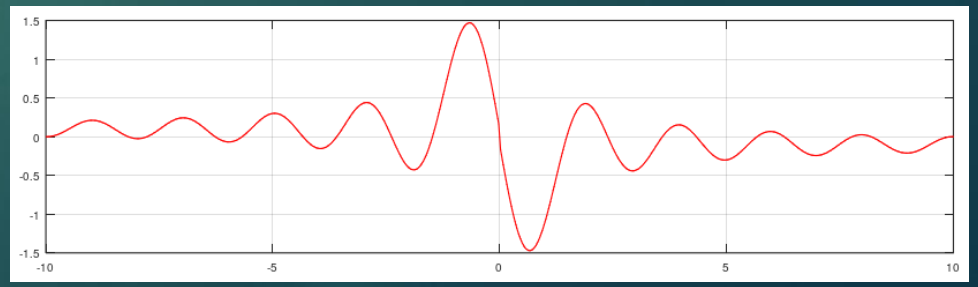
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_sc(t)+f_pt(t,20);
>> plot(t,x(t));
```




```
function y_out=dx(t,x)
```

$$x(t) \xrightarrow{y(t)=\frac{dx(t)}{dt}} y(t)$$

```
>> plot(t,dx(t,x),'r')
```



8) Sistema Integrador

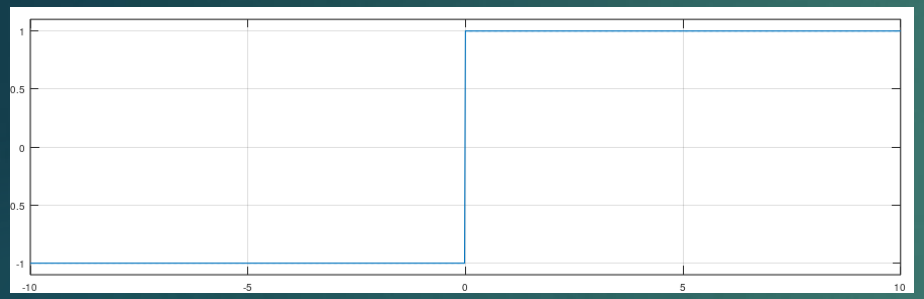
$$y(t) = \int_{-\infty}^{t} x(t)dt$$

$x(t)$ → Sinal de Entrada

$$\boldsymbol{H(s) = \frac{1}{s}}$$

$y(t)$ → Sinal de saída

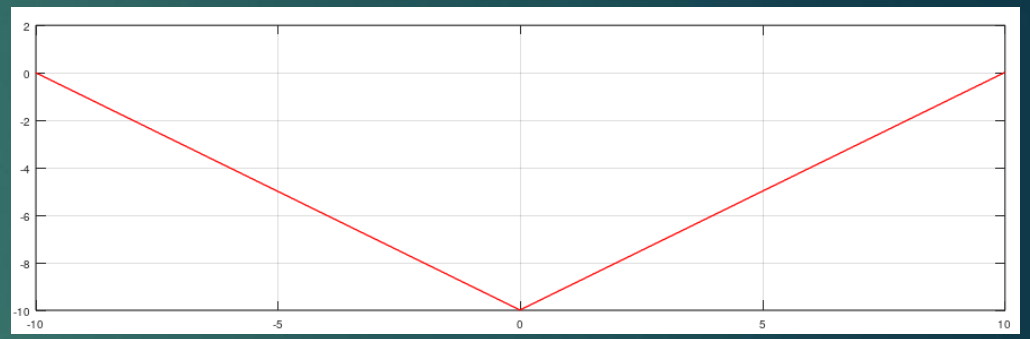
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_s(t);
>> plot(t,x(t));
```



```
function y_out=intx(t,x)
```

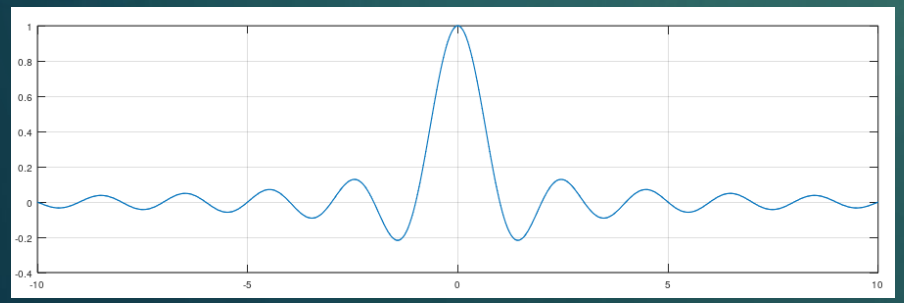
$$x(t) \xrightarrow{y(t)=\int_{-\infty}^{t} x(t)dt} y(t)$$

```
>> plot(t,intx(t,x),'r')
```



```
function y_out=intx(t,x)
    N=length(t);
    y_out(1)=0;
    for(n=2:N)
        y_out(n)=y_out(n-1) +
            (x(t(n))+x(t(n-1)))/2 * (t(n)-t(n-1));
    end
endfunction
```

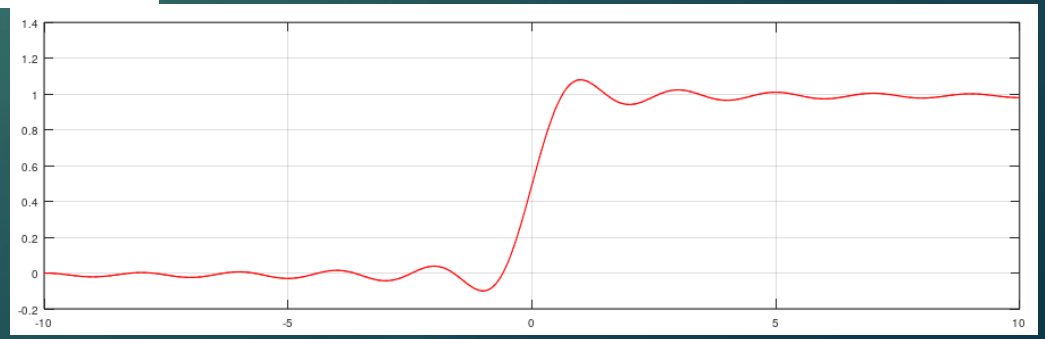
```
>> x = @(t) f_sc(t);
>> plot(t,x(t));
```



```
function y_out=intx(t,x)
```

$$x(t) \xrightarrow{y(t)=\int_{-\infty}^{t} x(t)dt} y(t)$$

```
>> plot(t,intx(t,x),'r')
```



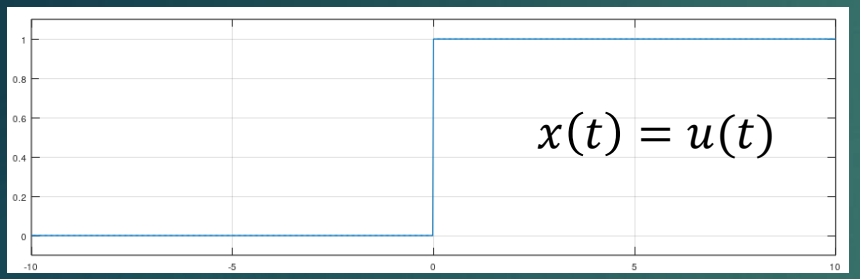
9) Sistema Descrito por uma Equação Diferencial (módulo 5)

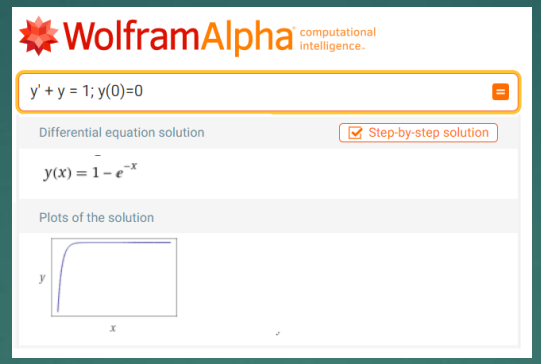
$$y'(t) + y(t) = x(t)$$

$$x(t) \xrightarrow{\text{Sinal de Entrada}} \boxed{H(s) = \frac{1}{s+1}} \xrightarrow{\text{Sinal de saída}} y(t)$$

Notar que a cada nova entrada, devemos resolver novamente a equação diferencial

```
>> t=linspace(-10,10,1001);
>> x = @(t) f_u(t);
>> plot(t,x(t));
```

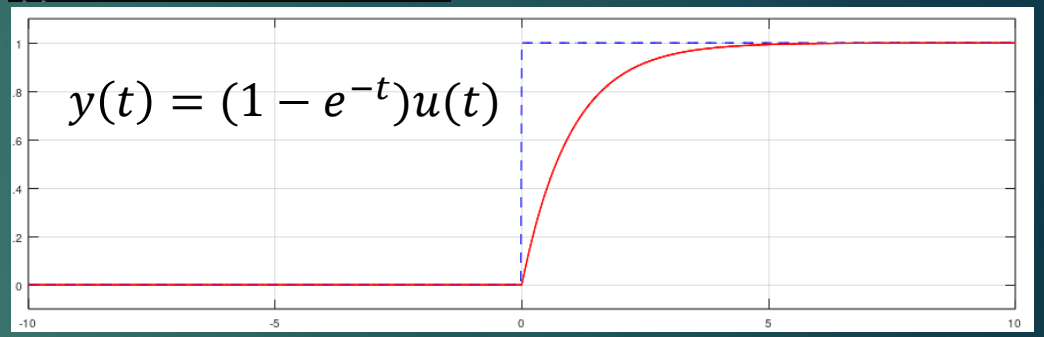




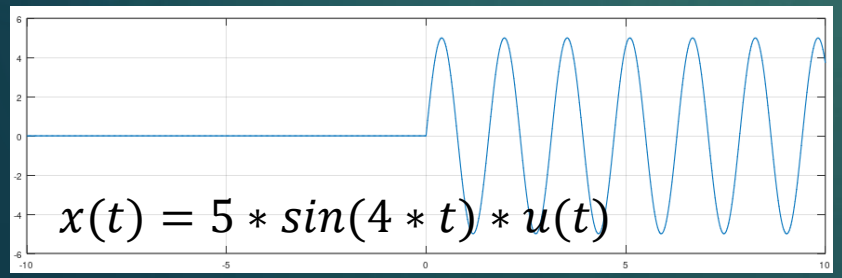
$$x(t) \xrightarrow[\text{Condição inicial } y(0)=0]{y'(t)+y(t)=x(t)} y(t)$$

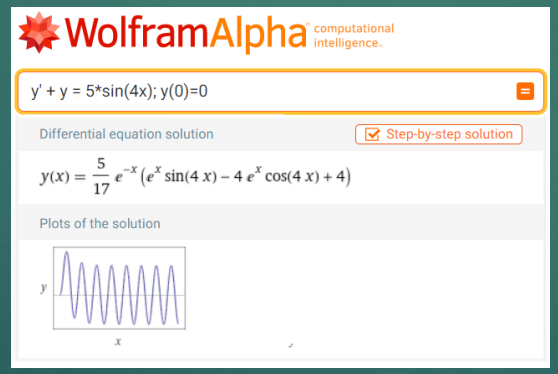
```
>> y = @(t) (1-exp(-t)).*f_u(t);
```

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y(t),'r');
```



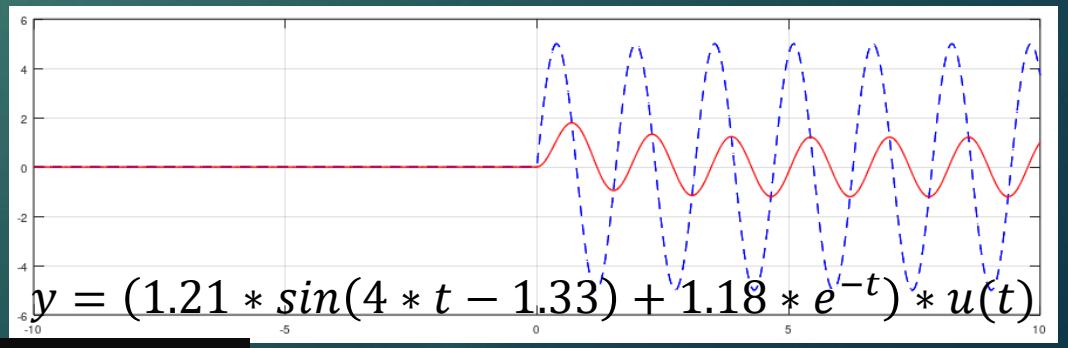

```
>> t=linspace(-10,10,1001);
>> x = @(t) 5*sin(4*t).*f_u(t);
>> plot(t,x(t));
```





$$x(t) \xrightarrow[\text{Condição inicial } y(0)=0]{y'(t)+y(t)=x(t)} y(t)$$

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y(t),'r');
```




```
>> y = @(t) (1.21*sin(4*t-1.33)+1.18*exp(-t)).*f_u(t);
```

10) Operador Atraso

$$y[n] = x[n-1]$$

$$x[n] \longrightarrow \boxed{\boldsymbol{Z^{-1}}} \longrightarrow y[n]$$

$$y[n] = x[n-1]$$

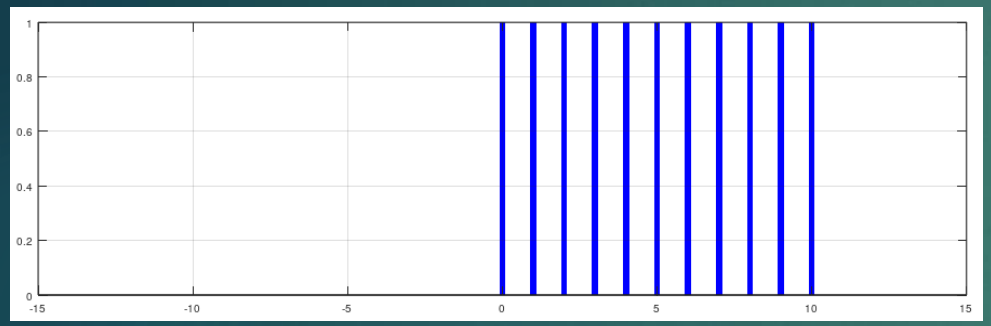
$$x[n] \longrightarrow \boxed{\boldsymbol{H(z)}} \longrightarrow y[n]$$

Sinal de Entrada

Sinal de saída

Sistema Discreto

```
>> n=[-10:10];
>> x = @(t) f_u(t);
>> bar(n,x(n),0.2,'b','edgecolor','none')
```

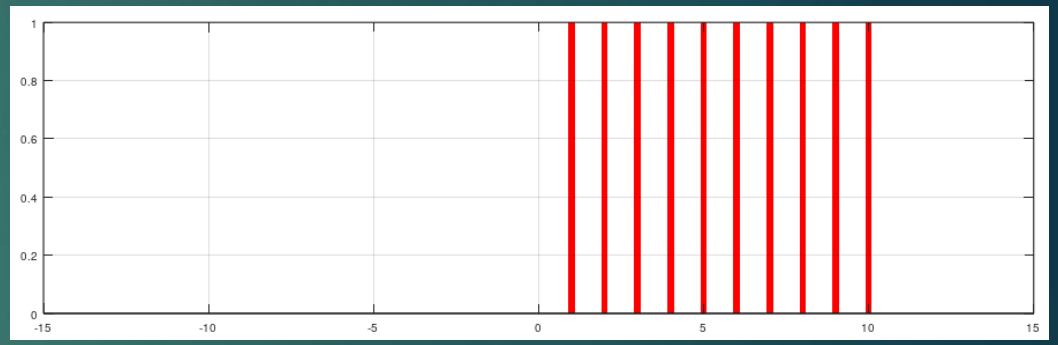


```
>> y1 = @(n,x) x(n-1);
```

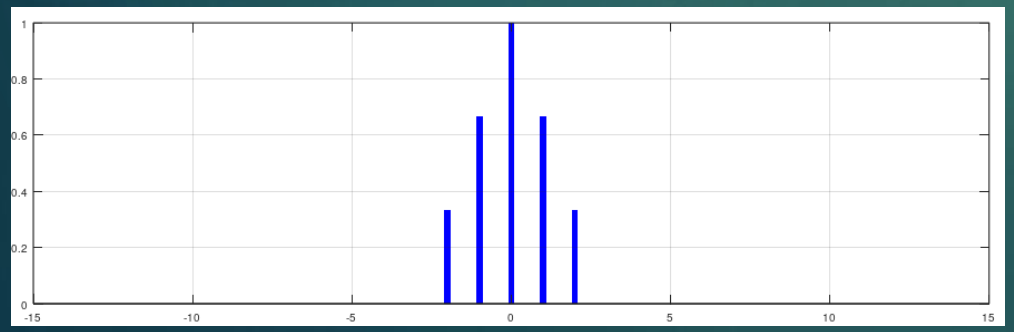
$$x[n] \xRightarrow{y1[n]=x[n-1]} y[n]$$

$$\boxed{\boldsymbol{H(z) = Z^{-1}}}$$

```
>> bar(n,y1(n,x),0.2,'r','edgecolor','none')
```



```
>> n=[-10:10];
>> x = @(t) f_pt(t,6);
>> bar(n,x(n),0.2,'b','edgecolor','none')
```

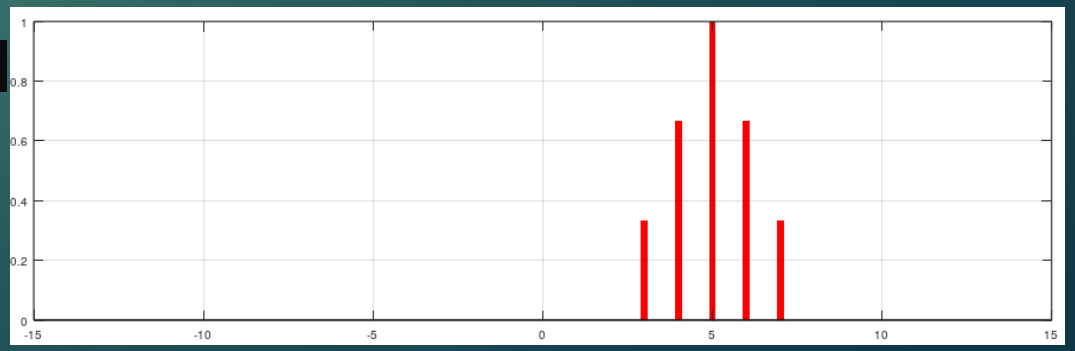


```
>> y2 = @(n,x) x(n-5);
```

$$x[n] \xRightarrow{y2[n]=x[n-5]} y[n]$$

$$\boxed{\boldsymbol{H(z) = Z^{-5}}}$$

```
>> bar(n,y2(n,x),0.2,'r','edgecolor','none')
```



11) Sistema Descrito por uma Equação de Diferenças

$$y[n] = 0.62\, y[n-1] + 0.38\, x[n]$$

$x[n]$

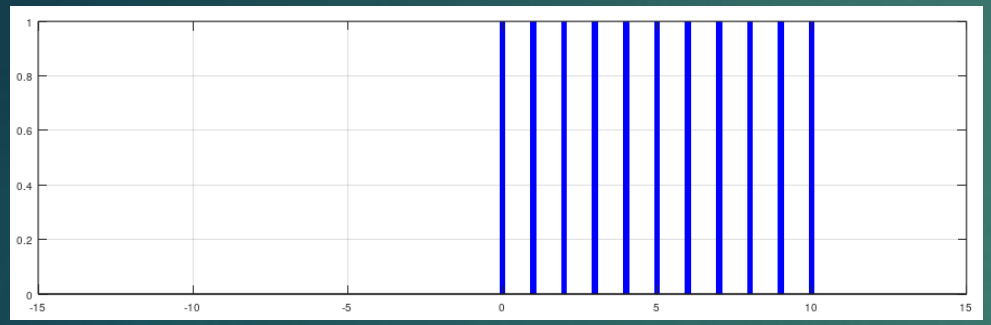
Sinal de Entrada

```
function y_out=rec1(t,x,y_ini)
    N=length(t);
    y_out=[y_ini];
    for (i=2:N)
        y_out(i)=0.62*x(t(i))+0.38*y_out(i-1);
    end
endfunction
```

$y[n]$

Sinal de saída

```
>> n=[-10:10];
>> x = @(t) f_u(t);
>> bar(n,x(n),0.2,'b','edgecolor','none')
```

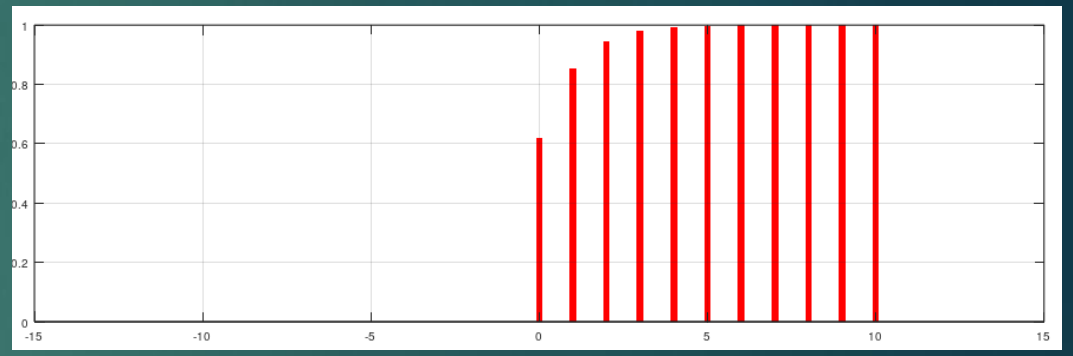


```
function y_out=rec1(t,x,y_ini)
```

$$x[n] \xrightarrow{y[n]=0.62\, y[n-1]+\, 0.38\, x[n]} y[n]$$

$$H(z) = \frac{0.38}{1 - 0.62z^{-1}}$$

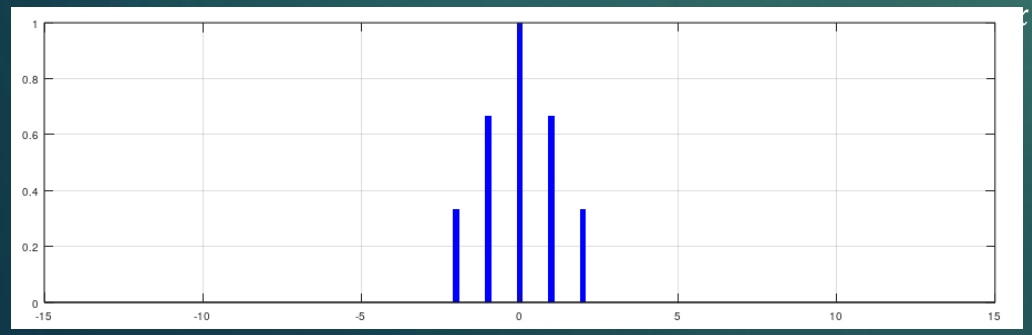
```
>> bar(n,rec1(n,x,0),0.2,'r','edgecolor','none');
```



Condição inicial

$y[0] = 0$

```
>> n=[-10:10];
>> x = @(t) f_pt(t,6);
>> bar(n,x(n),0.2,'b','edgecolor','none')
```



```
function y_out=rec1(t,x,y_ini)
```

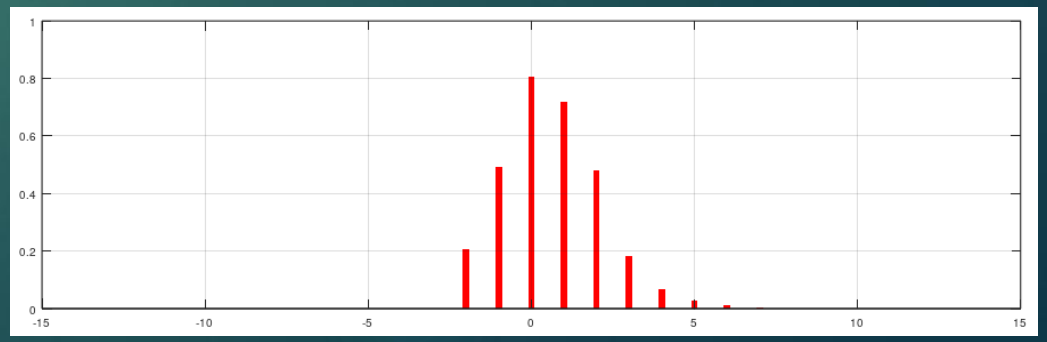
$$x[n] \xrightarrow{y[n]=0.62\, y[n-1]+\, 0.38\, x[n]} y[n]$$

$$H(z) = \frac{0.38}{1 - 0.62z^{-1}}$$

Condição inicial

$y[0] = 0$

```
>> bar(n,rec1(n,x,0),0.2,'r','edgecolor','none');
```



Propriedades de Sistemas LTI

Em Sinais e Sistemas descrevemos a relação entre o sinal de entrada e o sinal de saída através do conceito de função de transferência. As soluções que estudaremos neste curso são válidas para sistema LTI (Linear – time invariant), isto é sistemas lineares e invariantes no tempo.

Para entendermos melhor estes sistemas, iremos estudar algumas propriedades fundamentais, com destaque nas propriedades (1) e (2) que são necessárias em um sistema LTI:

1) **Linearidade**
2) **Invariância no tempo**
3) **Causalidade**
4) **Inversibilidade**
5) **Estabilidade**
6) **Memória**
7) **Realimentação**

1) Sistemas Lineares

Um sistema é Linear se e somente se satisfizer ao Princípio da Superposição, ou seja, satisfizer às propriedades de Homogeneidade e Aditividade. A resposta de um sistema linear a uma soma ponderada de sinais de entrada é igual a soma ponderada das saídas individuais de cada sinal de entrada.

Princípio da Homogeneidade

Se $x[n] \xRightarrow{H(z)} y[n]$

Então $a\,x[n] \xRightarrow{H(z)} a\,y[n]$

Princípio da Aditividade

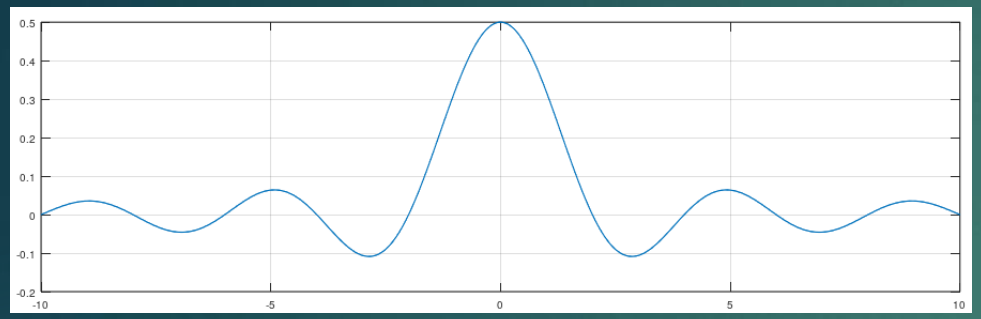
Se $x_1[n] \xRightarrow{H(z)} y_1[n]$

e $x_2[n] \xRightarrow{H(z)} y_2[n]$

Então $x_1[n] + x_2[n] \xRightarrow{H(z)} y_1[n] + y_2[n]$

Linear

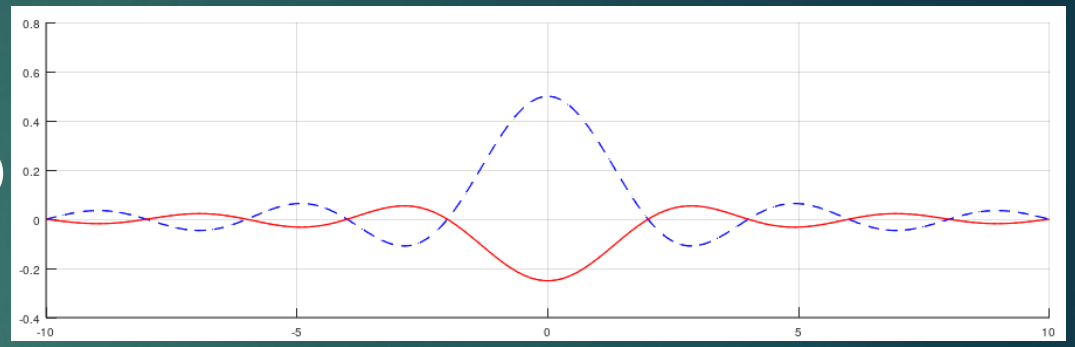
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_psc(t,2);
>> plot(t,x(t));
```




```
>> y1 = @(t,x) -0.5*x(t);
```

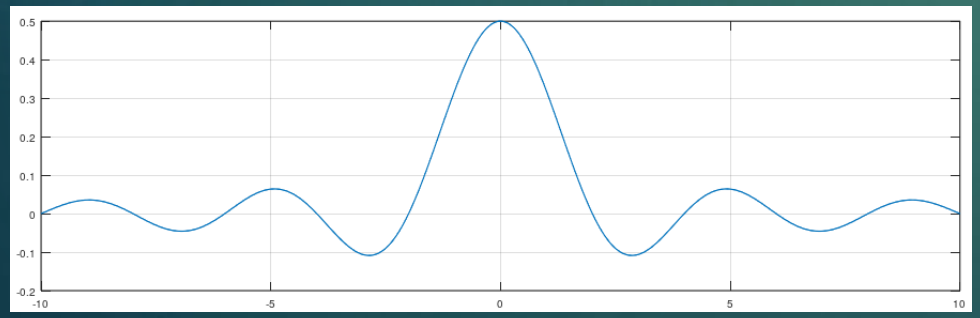
$$x(t) \xRightarrow{y(t)=-\frac{1}{2}x(t)} y(t)$$

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y1(t,x),'r');
```



Não-linear

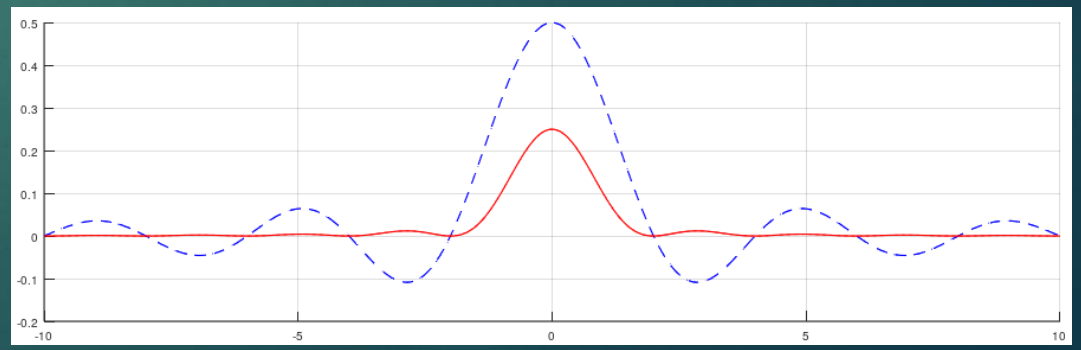
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_psc(t,2);
>> plot(t,x(t));
```




```
>> y2 = @(t,x) x(t).^2;
```

$$x(t) \xRightarrow{y(t)=x^2(t)} y(t)$$

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y2(t,x),'r');
```



2) Sistemas invariantes no tempo

- Em um sistema invariante no tempo, um atraso na sequencia de entrada causa um atraso equivalente na saída.

Se $x_1[n] \xRightarrow{H(z)} y_1[n]$

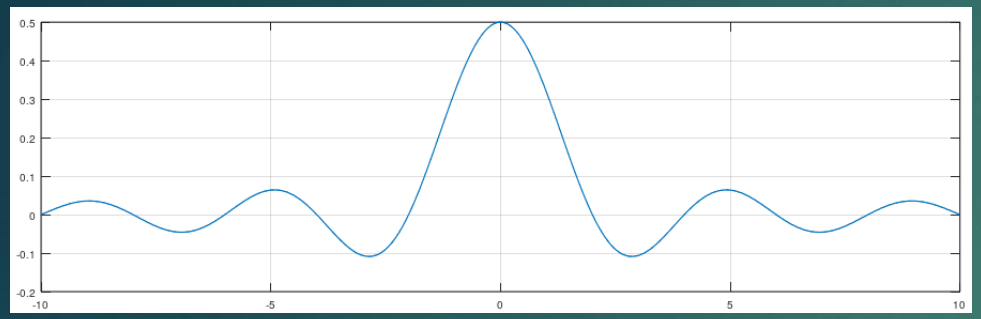
Então $x_2[n] = x_1[n-m] \xRightarrow{H(z)} y_2[n] = y_1[n-m]$

- Em um sistema variante no tempo, um atraso na sequencia de entrada causa mudanças na saída além de um simples atraso .

Invariante no Tempo

$$x(t) \xRightarrow{y(t)=-\frac{1}{2}x(t)} y(t)$$

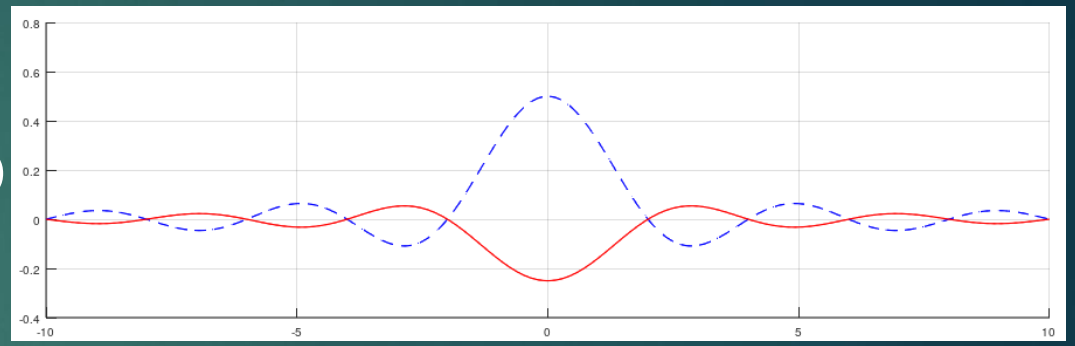
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_psc(t,2);
>> plot(t,x(t));
```



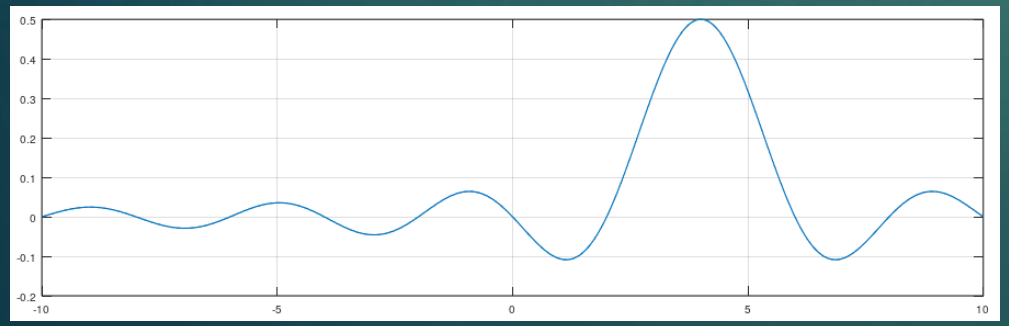

```
>> y1 = @(t,x) -0.5*x(t);
```

$$x(t) \xRightarrow{y(t)=-\frac{1}{2}x(t)} y(t)$$

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y1(t,x),'r');
```



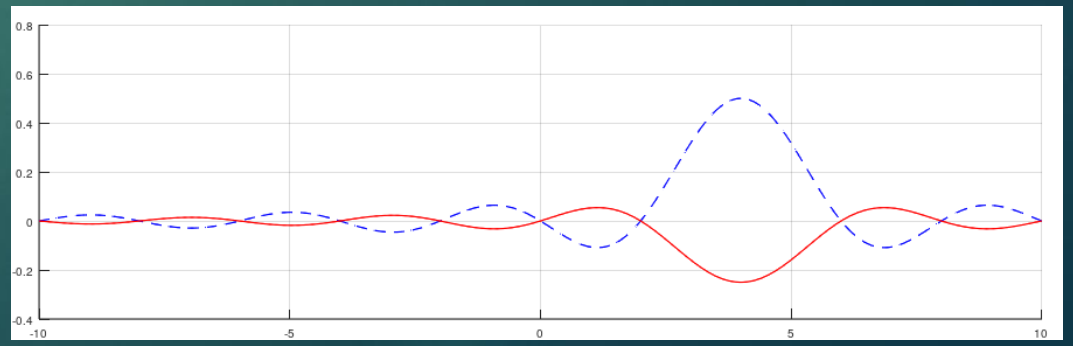

```
>> t=linspace(-10,10,1001);
>> x = @(t) f_psc(t-4,2);
>> plot(t,x(t));
```




```
>> y1 = @(t,x) -0.5*x(t);
```

$$x(t) \xRightarrow{y(t)=-\frac{1}{2}x(t)} y(t)$$

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y1(t,x),'r');
```



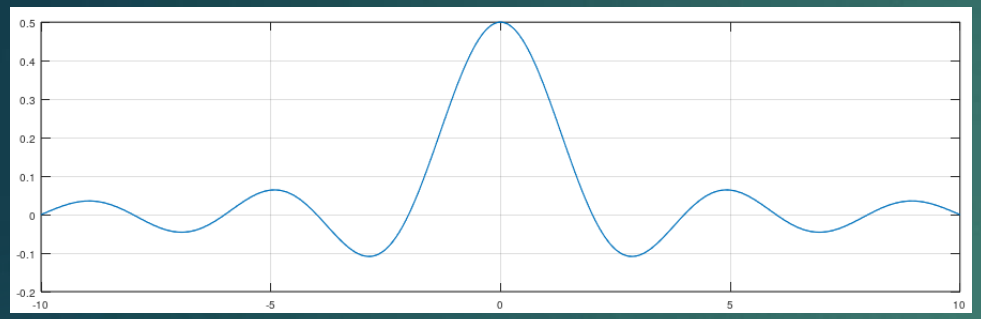
Variante no Tempo

$$x(t) \xRightarrow{y(t)=-\frac{t}{2}x(t)} y(t)$$

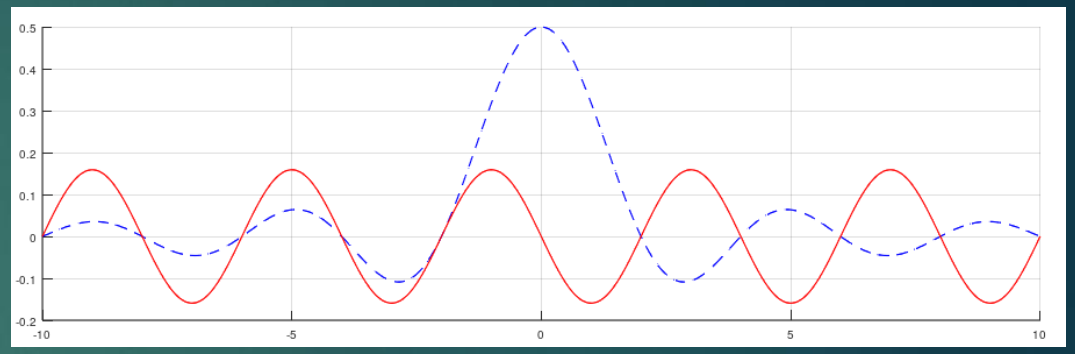
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_psc(t,2);
>> plot(t,x(t));
```

```
>> y2 = @(t,x) -1/2*t.*x(t);
```

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y2(t,x),'r');
```



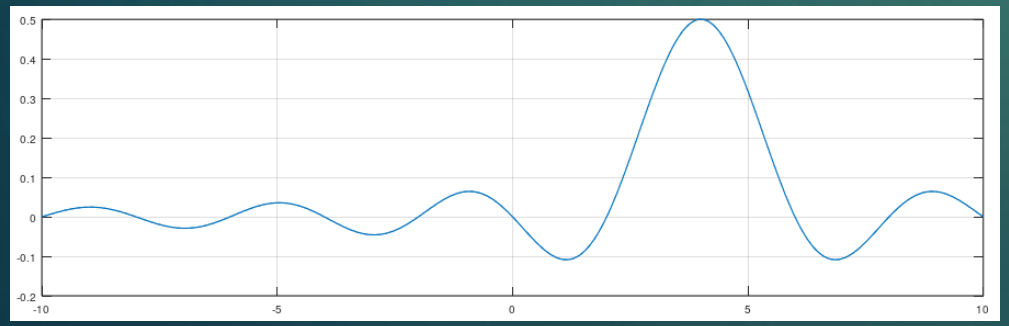
$$x(t) \xRightarrow{y(t)=-\frac{t}{2}x(t)} y(t)$$



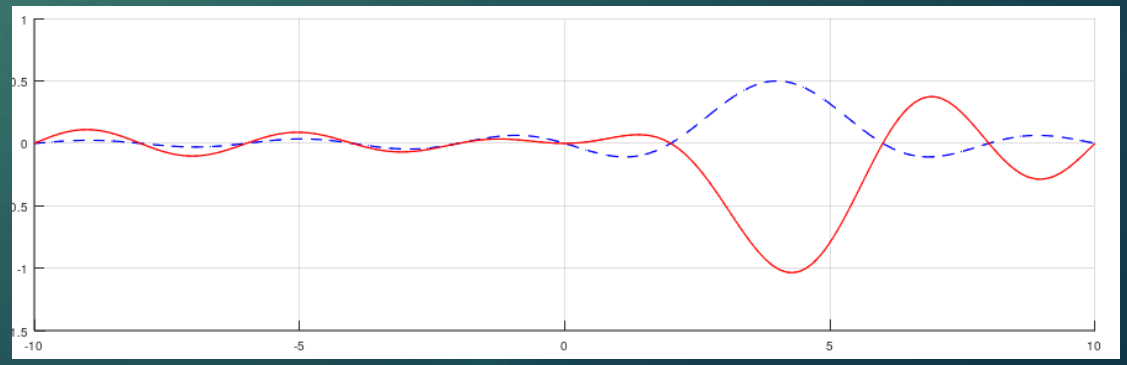
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_psc(t-4,2);
>> plot(t,x(t));
```

```
>> y2 = @(t,x) -1/2*t.*x(t);
```

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y2(t,x),'r');
```



$$x(t) \xRightarrow{y(t)=-\frac{t}{2}x(t)} y(t)$$



3) Sistemas causais

- Causalidade significa que a saída do sistema em um instante n não depende de nenhuma entrada que ocorra após n.
- Um sistema causal pode ser implementado em tempo real.
- Um sistema não-causal só em pós-processamento, quando as entradas já foram armazenadas.

$$x[n] \xRightarrow{H_1(z)} y_2[n] = 5x[n-2]$$

causal

Se $x_1[n] \xRightarrow{H(z)} y_1[n]$

e $x_2[n] \xRightarrow{H(z)} y_2[n]$

e $x_1[n] = x_2[n]$ para $\mathrm{n} \leq n_o$

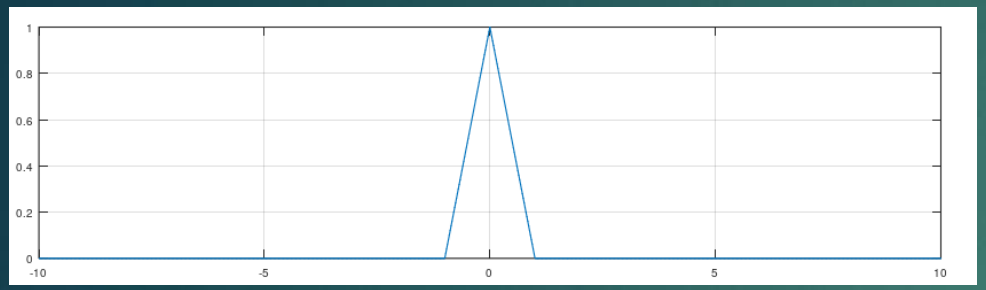
Então $y_1[n] = y_2[n]$ para $\mathrm{n} \leq n_o$

$$x[n] \xRightarrow{H_2(z)} y_2[n] = 2x[n+3]$$

Não -causal

Causal

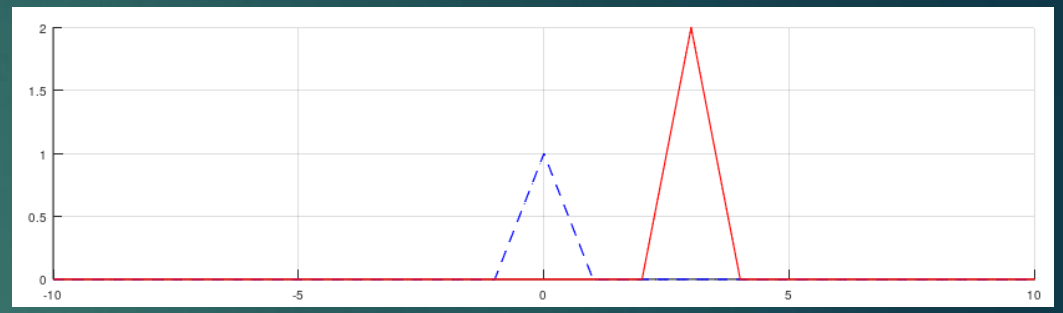
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_pt(t,2);
>> plot(t,x(t));
>>
```




```
>> y1 = @(t,x) 2*x(t-3);
```

$$x(t) \xRightarrow{y(t)=2x(t-3)} y(t)$$

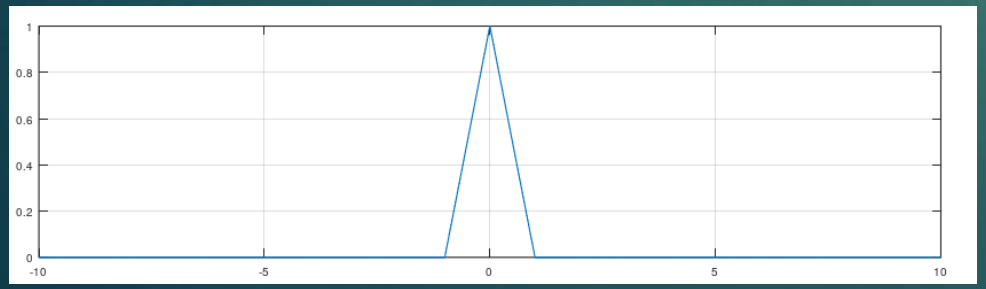
```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y1(t,x),'r');
```



Saída ocorre após a entrada

Não-causal

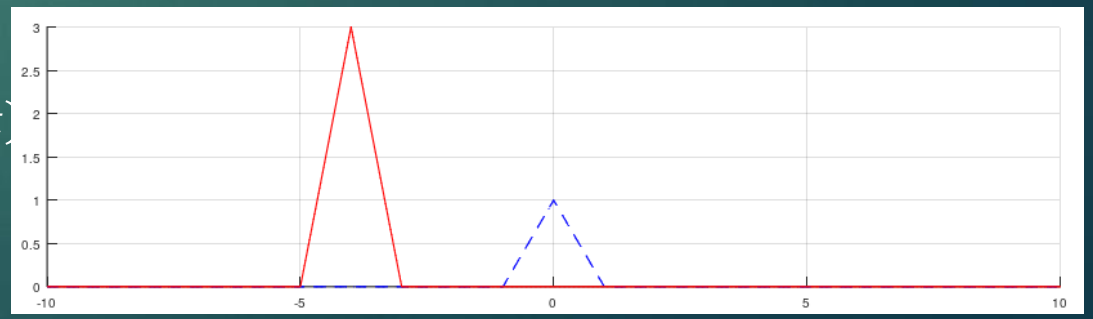
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_pt(t,2);
>> plot(t,x(t));
>>
```




```
>> y2 = @(t,x) 3*x(t+4);
```

$$x(t) \xRightarrow{y(t)=3x(t+4)} y(t)$$

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y2(t,x),'r');
```



Saída ocorre antes da entrada !!

Exemplos de linearidade, Variância no Tempo e Causalidade.

$$x[n] \xRightarrow{H(z)} y[n] = (n+5)x[n-4]$$

H(z) é linear, variante no tempo e causal

$$x[n] \xRightarrow{H(z)} y[n] = x^2[\text{n+1}]$$

H(z) é não-linear, invariante no tempo e não causal

4) Sistemas estáveis

A saída de um sistema estável não diverge se a entrada não divergir.

Entrada Limitada → Saída limitada (BIBO – *Bounded input Bounded output*)

Se $x(t) \xRightarrow{H(s)} y(t)$

E H(s) é estável Sistema contínuo

Então $|y(t)| < M_1 < \infty$ se $|x(t)| < M_2 < \infty$

Se $x[n] \xRightarrow{H(z)} y[n]$

E H(z) é estável Sistema Discreto

Então $|y[n]| < M_1 < \infty$ se $|x[n]| < M_2 < \infty$

- Em um sistema estável entradas limitadas produzam saídas limitadas (BIBO - Bounded Input Bounded Output).
- Nestes sistemas, entradas de baixa amplitude produzem saídas divergentes (que tendem a infinito).
- Basta que a soma dos módulos da resposta ao impulso do sistema seja limitada (ou integral da resposta ao impulso.

$$Se \quad x_1[n] \xRightarrow{H(z)} y_1[n] = \sum_{-\infty}^{\infty} x_1[n-k]h[k] \quad , onde\ h[n] \text{é}\ a\ resposta\ ao\ impulso\ do\ sistema$$

$$então \quad |y_1[n]| = \sum_{-\infty}^{\infty} |x_1[n-k]||h[k]|$$

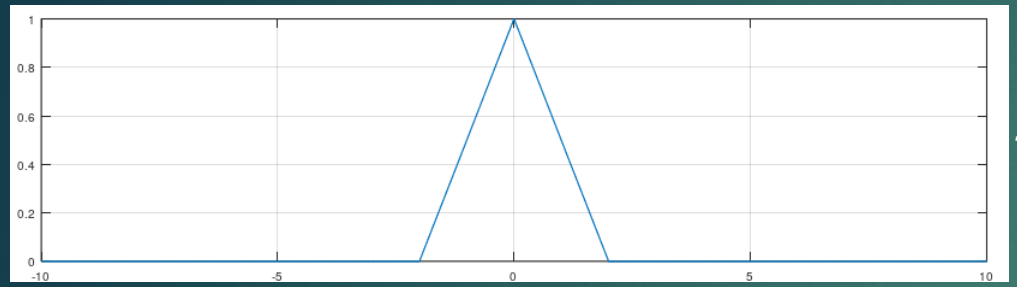
$$Como \quad x_1[n] \text{ for limitada temos que } |x_1[n]| = x_{max} < \infty$$

$$então \quad |y_1[n]| = x_{max} \sum_{k=-\infty}^{\infty} |h[k]|$$

$$Logo, \quad para \quad y_1[n] \text{ ser limitada basta que } \sum_{k=-\infty}^{\infty} |h[k])| \quad seja\ limitada$$

Estável

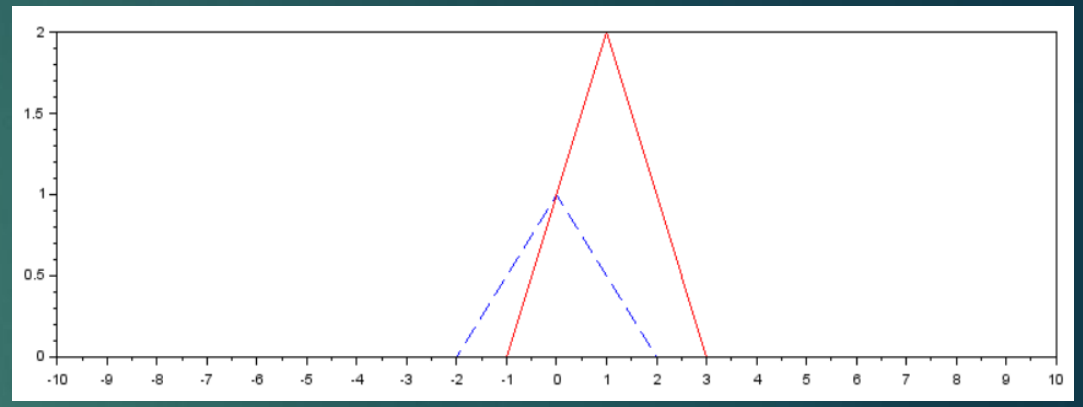
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_pt(t,4);
>> plot(t,x(t));
```



```
>> y1 = @(t,x) 2*x(t-1);
```

$$x(t) \xRightarrow{y(t)=2x(t-1)} y(t)$$

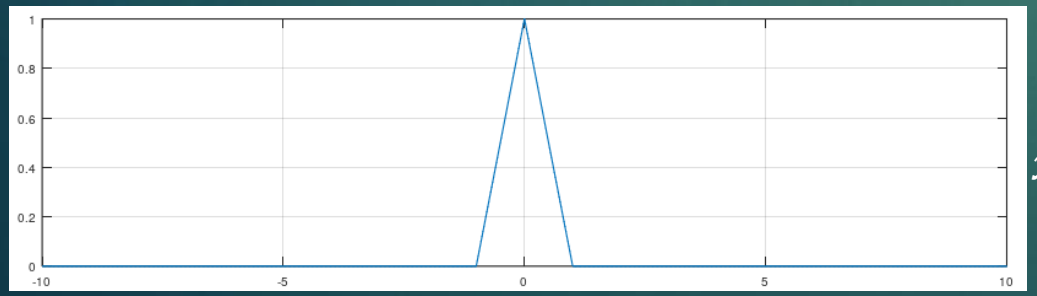
```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y2(t,x),'r');
```



Saída limitada, para entrada limitada

Não Estável

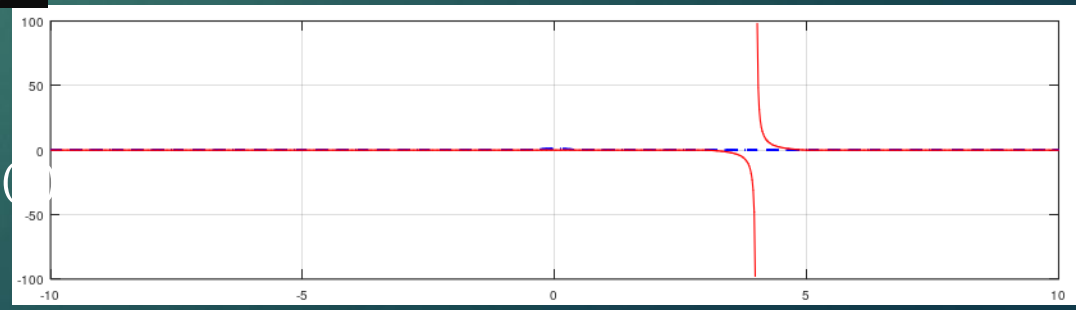
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_pt(t,2);
>> plot(t,x(t));
```



```
>> y2 = @(t,x) 2./(t-4).*x(t-4);
```

$$x(t) \xRightarrow{y(t)=\frac{2}{(t-4)}x(t-4)} y($$

```
>> hold on; grid on;
>> plot(t,x(t),'b--');
>> plot(t,y2(t,x),'r');
```



Em t=0 a saída encontra um polo (vai para ∞) embora a entrada seja limitada

5) Sistemas Inversíveis

Um sistema é inversível se:

Se $x(t) \xRightarrow{H(s)} y(t) \xRightarrow{H^{-1}(s)} x(t)$

Isto é, a entrada x(t) passa por um primeiro sistema H(s) gerando um saída y(t). Depois a saída y(t) se torna a entrada de um segundo sistema $H^{-1}(s)$, cuja saída é x(n) (idêntica à entrada do primeiro sistema)

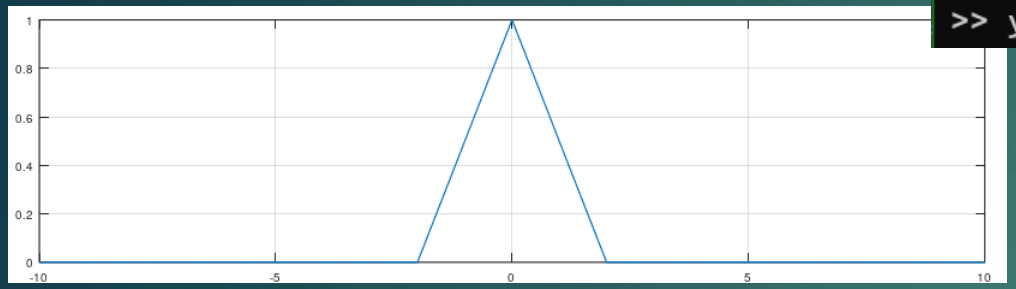
Exemplo: x(t) pode ser um sinal transmitido por um satélite. H(s) pode ser a distorção na ionosfera. y(t) seria então o sinal distorcido pela ionosfera recebido em um estação na terra. O sistemas $H^{-1}(s)$ pode ser então aplicado à y(n) para "reverter" a distorção e recuperar o sinal originalmente transmitido pelo satélite.

Sejam $H(s) = \frac{(s+4)(s+2)}{(s+5)(s+10)(s^2+8s+6)}$ e $H^{-1}(s) = \frac{(s+5)(s+10)(s^2+8s+6)}{(s+4)(s+2)}$

Então $H(s)H^{-1}(s) = 1$ Sistema Inversível

$$x(t) \xRightarrow{H(s)H^{-1}(s)} y(t) = \mathrm{x(t)}$$

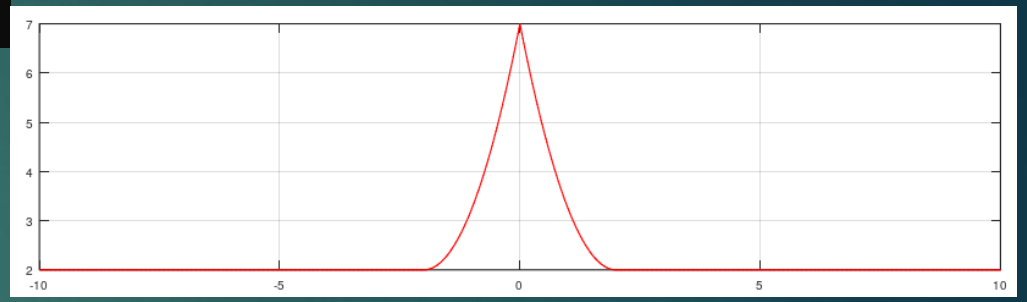
```
>> t=linspace(-10,10,1001);
>> x = @(t) f_pt(t,4);
>> plot(t,x(t));
```



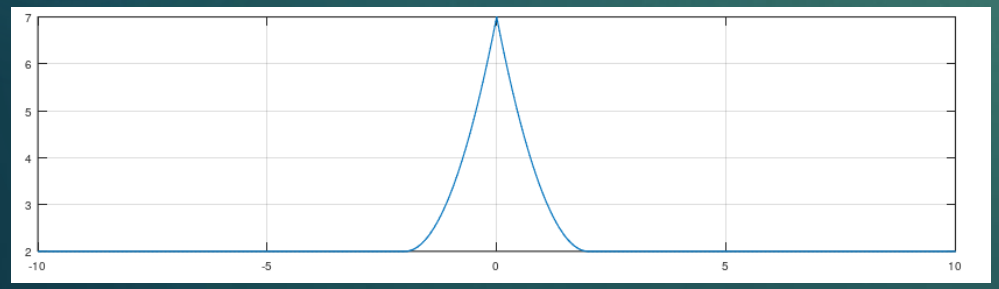
```
>> y_dir = @(t,x) 5*x(t).^2+2;
```

$$x(t) \xRightarrow{y(t)=5x^2(t)+2} y(t)$$

```
>> plot(t,y_dir(t,x),'r');
```



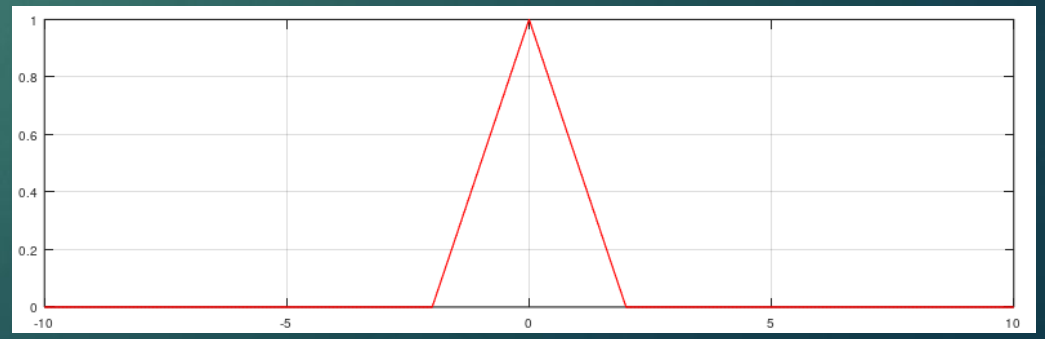
```
>> t=linspace(-10,10,1001);
>> x = @(t) 5*f_pt(t,4).^2+2;
>> plot(t,x(t));
```



```
>> y_inv = @(t,x) sqrt((x(t)-2)/5);
```

$$y(t) \xRightarrow{x_r=\sqrt{\frac{y(t)-2}{5}}} x_r(t)$$

```
>> plot(t,y_inv(t,x),'r');
```



Ao aplicarmos o sistema inverso, retornamos ao sinal original $x(t) = x_r(t)$

6) Sistemas com/sem memória

- Em um sistema sem memória a saída em determinado instante depende unicamente da entrada neste instante, isto é, não depende de entradas passadas ou futuras, ou saídas passadas ou futuras.
- Sistemas sem Memória também são conhecidos com Sistemas Estáticos
- $x(t) \xRightarrow{H(s)} 12\,x(t) + 4$
- $x[n] \xRightarrow{H(z)} 12\,x[n] + 4$
- Sistemas com memória dependem também de entrada passadas e/ou entradas futuras, e/ou de saídas passadas ou futuras.
- Sistema com memória são também conhecidos com Sistemas Dinâmicos.
- Sistemas com memória podem ser causais (dependem somente de entradas atual e passadas, e saídas passadas) ou anti-causais (dependem também de entradas/saídas futuras)
- $x(t) \xRightarrow{H(s)} \int_0^t (12\,x(t) + 4\,)dt$
- $x[n] \xRightarrow{H(z)} \frac{1}{3}(\,x[n] + 4\,x[n-1] - 3\,x[n-3])$
- $x[n] \xRightarrow{H(z)} -\,2x[n+1] + 5x[n-1] - 2\,x[n-3]$

7) Sistemas Diretos / Realimentados

Sistemas diretos só dependem das entradas atuais e/ou passadas e/ou futuras.

Sistemas realimentados dependem também (além das entradas) das saídas passadas

$y[n] = 2x[n+1] + 4x[n] + 3x[n-1] + 5x[n+2]$ Direto

$y[n] = y[n-1] + 3y[n-2] + 2x[n] + 7x[n-1] + 3x[n-2]$ Realimentado

$y(t) = 2 + 3x(t) - 7x'(t)$ Direto

$y(t) = y''(t) + 4y'(t) + 2x(t) + 3x'(t)$ Realimentado